

Linear Algebra

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Chapter 0

Prerequisites

0.1 Sets

Lecture 01: Sets

Definition (Set). Collection of objects where objects can be almost anything (number, symbol, set, shape...)

Note. This definition can lead to paradoxes, but its fine. Axiomatic set theories avoid this (e.g., Zermelo-Fraenkel), but usually the subtleties are not necessary.

- A set is described by the objects that **belong** to it (are **in** it)
- Sets are given (usually single, upper-cased, italicized, roman letter) names: A, B, X, P, R, T
- An object in a set is a **member** or **element** of A
- Belonging to (being a member of, being in) a set is denoted by $\in$ (e.g., $2 \in A$)

Definition 0.1.1 (Equal). Two sets, A and B are **equal** (denoted $A = B$) if every member of A is also a member of B and every member of B is also a member of A

Note. There is no order of members in a set (no “first,” or “last” member)

0.1.1 Set-builder notation

- To describe a small set, we can list members explicitly with curly braces, separated by commas (e.g., $A = \odot, \ominus, \spadesuit$)
- For larger (possibly infinite) sets we describe members using a predicate
 - A is the set of students in Quiz Bowl club
 - $\mathbb{N}$ is the set of Natural numbers
- The set-builder notation, $\{x : \Phi(x)\}$, is a concise expression of this
 - $A = \{x : x \text{ is a student in Quiz Bowl club}\}$
 - $B = \{x : x^2 = 4\}$
 - $C = \{2k : k \in \mathbb{N}\}$

Definition (Predicate). A logical formula that evaluates to True ($\top$) or False ($\perp$)

Definition (Domain of Discourse). Universe of objects that can potentially be in the set if they satisfy the predicate

Usually implied from the context, but can be explicitly defined:

$$E \in \mathbb{N} : (x \% 2) = 0\}$$

where $\mathbb{N}$ is the set of natural numbers (counting numbers):

$$\mathbb{N} = 1, 2, 3, \dots$$

0.1.2 Logic

Definition (Conditional Operator). Denoted $p \Rightarrow q$ (“if p then q ” or “ p implies q ”)

$$(p \Rightarrow q) = \begin{cases} \perp & \text{if } p = \top, q = \perp \\ \top & \text{otherwise} \end{cases}$$

Definition (Conjunction Operator). Denoted $p \wedge q$ (“ p and q ”)

$$(p \wedge q) = \begin{cases} \top & \text{if } p = \top, q = \top \\ \perp & \text{otherwise} \end{cases}$$

Definition (Disjunction Operator). Denoted $p \vee q$ (“ p or q ”)

$$(p \vee q) = \begin{cases} \perp & \text{if } p = \perp, q = \perp \\ \top & \text{otherwise} \end{cases}$$

Definition (Negation Operator). Denoted $\neg p$ (“not p ”)

$$(\neg p) = \begin{cases} \top & p = \perp \\ \perp & p = \top \end{cases}$$

Definition (Biconditional Operator). Denoted $p \Leftrightarrow q$ (“ p if and only if q ”)

$$(p \Leftrightarrow q) = (p \Rightarrow q) \wedge (q \Rightarrow p)$$

0.1.3 Set Notation and Terminology

Let A, B be sets from the same [domain of discourse](#)

Definition 0.1.2 (Subset). A is called a **subset** of B , denoted $A \subseteq B$, if

$$(x \in A) \Rightarrow (x \in B)$$

Note.

$$A \subseteq A$$

Definition 0.1.3 (Intersection). The **intersection** of A and B , denoted $A \cap B$, is the set

$$(A \cap B) = \{x : (x \in A) \wedge (x \in B)\}$$

Definition 0.1.4 (Union). The **union** of A and B , denoted $A \cup B$, is the set

$$(A \cup B) = \{x : (x \in A) \vee (x \in B)\}$$

Notation. The **empty set**, denoted $\emptyset$, contains no members

Note.

$$\emptyset \subseteq A$$

Definition 0.1.5 (Power Set). Let A be a set; the **power set of** A , denoted $P(A)$, is the set of all subsets of A :

$$P(A) = \{S : S \subseteq A\}$$

Example.

$$A = \{\odot, \ominus, \spadesuit\}$$
$$P(A) = \{\emptyset, \{\odot\}, \{\ominus\}, \{\spadesuit\}, \{\odot, \ominus\}, \{\odot, \spadesuit\}, \{\ominus, \spadesuit\}, \{\odot, \ominus, \spadesuit\}\}$$

Definition 0.1.6 (Cartesian Product). Let A, B be sets; the **Cartesian product** of A with B , denoted $A \times B$ is the set of all ordered pairs of items, the first taken from A and the second taken from B

$$A \times B = \{(x, y) : x \in A, y \in B\}$$

Example.

$$A = \{\odot, \ominus, \spadesuit\}, B = \{\circ, \diamond\}$$
$$A \times B = \{(\odot, \circ), (\odot, \diamond), (\ominus, \circ), (\ominus, \diamond), (\spadesuit, \circ), (\spadesuit, \diamond)\}$$

Note. We denote an ordered pair with parenthesis, not curly braces

- (x, y) is not the same as x, y because order matters
- $x, y = y, x$ but $(x, y) \neq (y, x)$

0.2 Proofs

Lecture 02: Proofs

Definition (Deductive Reasoning). The process of making deductive arguments

Definition (Deductive Argument). Process of making a logical inference

Definition (Inference). Claim that a certain [predicate](#), called the **conclusion**, follows from one or more [predicates](#), called the **premises**.

Predicate B **follows from** A if it is impossible simultaneously for A to be True and B to be False

An inference is **valid** if the conclusion follows from the premises

A deductive argument is **sound** if the inference is valid and its premises are True

Definition (Universal Quantification Symbol ($\forall$)). Denotes that a proposition is True for all members

Example.

$$\forall x \in \mathbb{N} : x^2 \geq x$$

Read: “**for all** (every, any, each) x in $\mathbb{N}$ the predicate $(x^2 \geq x)$ evaluates to True”

Definition (Existential Quantification Symbol($\exists$)). Denotes that a proposition is True for at least one member

Example.

$$\exists x \in \mathbb{N} : x^2 < x$$

Read: “**there exists** an x in $\mathbb{N}$ such that the predicate $x^2 < x$ evaluates to true”

Note. This is an example of a False proposition

0.2.3 Contraposition

Definition (Contraposition). Let p, q be predicates and consider the conditional statement

$$p \Rightarrow q$$

The **contrapositive** form of the statement is

$$\neg q \Rightarrow \neg p$$

A conditional statement and its contrapositive form are **equivalent**

$$(p \Rightarrow q) \Leftrightarrow (\neg q \Rightarrow \neg p)$$

When judging truthfulness of a statement, it sometimes helps to consider its contrapositive

Also existing, but not as common are:

- The **inverse**: $\neg p \Rightarrow \neg q$
- The **converse**: $q \Rightarrow p$
- The **complement**: $\neg(p \Rightarrow q)$

0.2.4 Proofs

Definition (Mathematical Proof). A deductive argument about something related to math. Uses spoken/written language, or even sketches/diagrams.

Usually rigorous (spells out assumptions and deductive steps as is convenient) but informal (some natural language with occasionally ambiguous symbols/rules) deductive reasoning

Proposition 0.2.1. Let $n, m \in \mathbb{N}$ and suppose n, m are even; then $(n + m)$ is even

Proof.

- Since n is even $\exists k \in \mathbb{N}$ such that $n = 2k$
- Since m is even $\exists q \in \mathbb{N}$ such that $m = 2q$
- Then $(n + m) = (2k + 2q) = 2(k + q)$
- Therefore, $(n + m)$ is even

Note. We used a method called **direct proof**

0.2.5 Mathematical Induction

Definition (Mathematical Induction). If asked to prove that a certain proposition, $P(n)$ is true for any $n \in \mathbb{N}$, we will accept as proof the following inference, if sound:

$$\left(P(k) \stackrel{(1)}{=} \text{T} \right) \Rightarrow (P(k+1) = \text{T}) \quad \stackrel{(2)}{\Rightarrow} P(n) = \text{T}, \forall n \in \mathbb{N} \quad (1)$$

Where (1) is called the **base case**; (2) is called the **induction hypothesis**; (3) is the **induction step**

A variant called **complete** (or strong/generalized) induction uses a stronger hypothesis in the induction step:

$$(P(j), \forall j \leq k) \Rightarrow P(k+1)$$

Proposition 0.2.2. For any $n \in \mathbb{N}$

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

Proof. By induction

- Base case: for $n = 1$, $\frac{n(n+1)}{2} = 1$
- Induction step: suppose, for some $k \in \mathbb{N}$, $1 + 2 + \cdots + k = \frac{k(k+1)}{2}$, then

$$\begin{aligned} 1 + 2 + \cdots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1)}{2} + \frac{2(k+1)}{2} = \frac{(k+1)(k+2)}{2} \end{aligned}$$

Proposition 0.2.3. Let $n \in \mathbb{N}$, if n^2 is even then n is even

Proof. By Contraposition
Suppose n were not even, then

- n is odd, therefore
- $\exists k \in \mathbb{N}$ such that $n = 2k - 1$, and therefore
- $n^2 = n \cdot n = (2k - 1)(2k - 1) = 4k^2 - 4k + 1 = 4(k^2 - k) + 1$
- Let $p = 4(k^2 - k)$, then
- $p \div 2 = 2(k^2 - k)$, therefore
- p is even, therefore
- $n^2 = p + 1$ is odd
- Since n^2 is not odd, n cannot be not even

Proposition 0.2.4. $\sqrt{2}$ is irrational

Proof. By contradiction
Suppose $\sqrt{2}$ is rational, then

- $\exists p, q \in \mathbb{N}$ such that $\sqrt{2} = \frac{p}{q}$ and p and q have no common factors,

0.2.6 Things that often need proving

- Existence: $\exists x$ such that $P(x)$ Proof by construction describes who the required x is, and verifies $P(x) = \text{T}$ (Don't need to actually find the thing)
- Uniqueness: $(P(x) \wedge P(y)) \Rightarrow (x = y)$ Name two things with the property and show they must be equal

0.2.7 Proof of Equivalence

To prove $P \Leftrightarrow Q$, prove both $P \Rightarrow Q$ and $Q \Rightarrow P$ Related: show two sets A, B are equal by showing $A \subseteq B$ and $B \subseteq A$ To prove the following are equivalent (TFAE) you could prove a circular chain of implications

$$(A \Rightarrow B) \wedge (B \Rightarrow C) \wedge (C \Rightarrow D) \wedge (D \Rightarrow A)$$

0.3 Operations

Lecture 03: Operations

0.3.1 Function

Definition 0.3.1 (Function). A function comprises three objects:

- A set, called the **domain**
- Another set, called the **range**
- A mapping of each member of the domain to a single member of the range, called its **image**

Notation. Often we use lower-case Latin letters to name functions.

The notation $f : D \rightarrow R$ reads “ f is a function from the domain set D into the range set R ”.

The image of $x \in D$ is denoted $f(x)$ (read “ f of x ”).

Definition 0.3.2 (Image of a Set). Let $f : D \rightarrow R$ be a function and let $U \subseteq D$.

The set of images of members of U is called the **image of** U , and denoted $f(U)$:

$$f(U) = \{f(x) : x \in U\}$$

The set of images of all members of the domain, $f(D)$, is called the **image of the function**.

Example. Let $D = \{-2, 2, 3, 5, 7\}$ and let $h : D \rightarrow \mathbb{N}$ be the mapping

$$h(-2) = 4, \quad h(2) = 4, \quad h(3) = 9, \quad h(5) = 25, \quad h(7) = 49$$

The domain and range are given explicitly. The image is evident from the mapping; the image of h is the set

$$S = \{4, 9, 25, 49\}$$

With D and S as above, let $f : D \rightarrow S$ be the mapping

$$f(-2) = 4, \quad f(2) = 4, \quad f(3) = 9, \quad f(5) = 25, \quad f(7) = 49$$

Strictly speaking, $f : D \rightarrow S$ and $h : D \rightarrow \mathbb{N}$ are different functions (they have different ranges). This distinction is not that important.

0.3.2 Ways to Describe a Function

There are many different ways to describe a function:

- **A table:**

Domain	-2	2	3	5	7
Image	4	4	9	25	49

- **A formula:** $f(x) = x^2$
- **A set of ordered pairs:** $f = \{(-2, 4), (2, 4), (3, 9), (5, 25), (7, 49)\}$
- **A set of ordered pairs in set builder notation:**

$$f = \{(x, y) : x \in \{-2, 2, 3, 5, 7\}, y = x^2\}$$

Note. Not every set of ordered pairs describes a valid function. For example:

$$\{(2, 2), (2, 3), (3, 1)\}$$

is not a function because the element 2 in the domain maps to two different values.

0.3.3 Bijection

Definition 0.3.3 (Injective, Surjective, Bijective). Let $f : D \rightarrow R$ be a function.

- (i) f is called **injective** (or **one-to-one**) if distinct members of D are mapped to distinct members of R :

$$\forall x, x' \in D, \quad x \neq x' \Rightarrow f(x) \neq f(x')$$

- (ii) f is called **surjective** (or **onto**) if every element of R is the image of some element of D :

$$\forall y \in R, \quad \exists x \in D : y = f(x)$$

- (iii) f is called **bijective** if it is both injective and surjective (one-to-one and onto).

Note. Students are sometimes confused about the surjective part because, in calculus, it is customary to let the range of a function be implicitly defined as equal to its image, making every function surjective.

0.3.4 Binary Operations

Remember the four basic arithmetic operations? What they have in common is that they each take two numbers as input and produce one number as output.

Definition 0.3.4 (Binary Operation). Let A be a set. A **binary operation** on A is a function whose domain is the **Cartesian Product** $A \times A$.

Notation. An operation is often denoted with a symbol, like $\star$, instead of a letter, and the image of a pair from the domain is denoted with the operator between the pair items:

$$x \star y \quad \text{instead of} \quad f(x, y)$$

Example. Some examples of binary operations $\star$ on $\mathbb{N}$ are:

- $a \star b := a + b + 8$
- $a \star b := \max(a, b)$

- $a \star b :=$ the digit in the ones place of $a + b^2$

0.3.5 Range of Operations and Closure

The sum and product of any two natural numbers is a natural number:

$$\begin{aligned} \forall a, b \in \mathbb{N}, \quad a + b \in \mathbb{N} \\ \forall a, b \in \mathbb{N}, \quad ab \in \mathbb{N} \end{aligned}$$

This is not true for subtraction and division:

$$\begin{aligned} \exists a, b \in \mathbb{N} : a - b \notin \mathbb{N} \\ \exists a, b \in \mathbb{N} : a/b \notin \mathbb{N} \end{aligned}$$

This convenient property of addition and multiplication is called **closure**.

Definition 0.3.5 (Closure). A set A is said to be **closed with respect to an operation $\star$** if

$$\forall a, b \in A, \quad (a \star b) \in A$$

Example.

- The set $\mathbb{N}$ is closed with respect to addition and with respect to multiplication
- The set $\mathbb{N}$ is not closed with respect to subtraction
- The set $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ is closed with respect to subtraction

0.3.6 Associativity

Definition 0.3.6 (Associative Operation). An operation $\star$ on a set A is called **associative** if, $\forall a, b, c \in A$:

$$(a \star b) \star c = a \star (b \star c)$$

Example.

- Addition and multiplication are associative operations on $\mathbb{Z}$
- Subtraction is not associative on $\mathbb{Z}$

Note. If $\star$ is an associative operation on A , it is customary to neglect parentheses:

$$(a \star b) \star c = a \star (b \star c) = a \star b \star c$$

It is implied in the definition of associativity that the set A is closed with respect to $\star$.

Problem (CFU). If $\star$ is associative on A , is it true that $\forall a, b, c, d \in A$:

$$(a \star b) \star (c \star d) = a \star (b \star c) \star d$$

Bonus CFU: can you prove your answer?

Answer. Yes.

$$(a \star b) \star (c \star d) = ((a \star b) \star c) \star d = (a \star (b \star c)) \star d = a \star (b \star c) \star d$$

⊗

Theorem 0.3.7 (Generalized Associativity). Let $\star$ be an associative operation on a set F . Then, for any $a_1, a_2, \dots, a_n \in F$, every possible parenthesis ordering of

$$a_1 \star a_2 \star a_3 \star \dots \star a_{n-1} \star a_n$$

is equivalent to the left-associated ordering:

$$(\dots((a_1 \star a_2) \star a_3) \star \dots \star a_n)$$

Proof. By generalized (strong) induction.

Base case: For $n = 1$ or $n = 2$ there is nothing to prove, and for $n = 3$, there are two possible orderings, and regular associativity guarantees that $(a_1 \star a_2) \star a_3$ is equal to the left-associated ordering $((a_1 \star a_2) \star a_3)$.

Inductive step: For $n > 3$, we suppose that every parenthesis ordering of an expression with fewer than n operands is equivalent to the left-associated ordering, and proceed to consider the expression with n operands:

$$E = a_1 \star a_2 \star \dots \star a_n$$

Any ordering of parentheses ends with a final step:

$$E = L \star R$$

where L is some parenthesis ordering of $a_1 \star \dots \star a_q$ and R is some parenthesis ordering of $a_{q+1} \star \dots \star a_n$, and $q < n$.

By the induction hypothesis, both L and R are equal to their respective left-associated orderings:

$$\begin{aligned} L &= (\dots((a_1 \star a_2) \star \dots) \star a_q) \\ R &= (\dots((a_{q+1} \star a_{q+2}) \star \dots) \star a_n) \end{aligned}$$

If $q = n - 1$, so that $R = a_n$, then $E = L \star R$ is already left-associated. Otherwise, write

$$R = M \star a_n = (\dots((a_{q+1} \star a_{q+2}) \star \dots) \star a_{n-1}) \star a_n$$

and

$$E = L \star (M \star a_n) = (L \star M) \star a_n$$

by regular associativity. By the induction hypothesis, since $L \star M$ includes $n - 1$ operands, it is equivalent to the left-associated ordering of $a_1 \star \dots \star a_{n-1}$, making $E = (L \star M) \star a_n$ the left-associated ordering of the original expression. ■

0.3.7 Neutrality

Definition 0.3.8 (Neutral Element). Let $\star$ be an operation on the set A . An element $e \in A$ is called **neutral with respect to $\star$** if

$$\forall a \in A, \quad a \star e = e \star a = a$$

Theorem 0.3.9 (Uniqueness of the Neutral). Let $\star$ be an operation on a set A . There is at most one element in A that is neutral with respect to $\star$.

Proof. Suppose $p, q \in A$ are both neutral with respect to $\star$. In that case:

- $p \star q = q$, because p is neutral
- $p \star q = p$, because q is neutral
- Therefore, $p = q$

■

Note. Since a neutral element in A is unique, if it exists, we can call it **the** neutral.

Problem (Exercise). Let $\star$ denote the operation on $\mathbb{N}$ defined by $a \star b := a^b$.

- Is there, in $\mathbb{N}$, a neutral with respect to $\star$?
- Can you prove your answer?

Answer.

- No
- Yes:
 - If $a^x = a, \forall a \in \mathbb{N}$, then in particular $2^x = 2$ and therefore $x = 1$
 - So, the only candidate for being a neutral with respect to $\star$ is 1
 - However, if 1 is neutral with respect to $\star$, then $1^a = a, \forall a \in \mathbb{N}$, and in particular $1^2 = 2$
 - Which it is not
 - So there is no neutral element

⊗

0.3.8 Invertible Elements

Definition 0.3.10 (Invertible Element). Let $\star$ be an operation on A and let $e \in A$ be the neutral with respect to $\star$. An element $a \in A$ is called **invertible** if there exists an element $b \in A$ such that

$$\begin{aligned} a \star b &= e \\ b \star a &= e \end{aligned}$$

If such an element b exists, it is called **an inverse of a** .

Example.

- Every element of $\mathbb{Z}$ is invertible with respect to addition
- In $\mathbb{N}$, 1 is the only invertible element with respect to multiplication
- In $\mathbb{Z}$, 1 and -1 are the only invertible elements with respect to multiplication

Chapter 1

Fields

1.1 Definition

Lecture 04: Definition

1.1.1 Algebra

Algebra is the study of equations and calculations. Calculations are done by carrying out operations on objects of an **algebraic structure**.

Definition (Algebraic Structure). A set of objects with operations defined on it.

1.1.2 Numbers as Models

Numbers are an algebraic structure constructed to model certain types of physical objects and their interactions:

- Objects such as apples and skittles
- Interactions such as trading skittles for apples

Interactions in the natural world have certain characteristics, discovered by observation. Operations on numbers have properties matching those observations.

1.1.3 The Natural Numbers Model

The natural numbers $\mathbb{N}$ consist of:

- A set $\mathbb{N}$ of objects, called numbers
- An algorithm assigning each number a unique name
- An algorithm for deciding which number is "next"
- A binary operation called addition, denoted by $+$
- An algorithm for finding $a + b$ for any $a, b \in \mathbb{N}$

These algorithms are constructed to have properties modeled after observed characteristics of combining physical objects.

1.1.4 The Integer Numbers Model

Additional observations reveal interactions not adequately modeled by $\mathbb{N}$. The integers $\mathbb{Z}$ extend $\mathbb{N}$ by:

- Adding a new number, called zero and denoted 0
- Matching each number $a \in \mathbb{N}$ with a new number $-a$
- Extending the addition algorithm for these new numbers

1.1.5 Abstraction and Generalization

Exploration in mathematics proceeds by abstraction and generalization:

1. Start with a familiar mathematical model whose properties are understood
2. Investigate a hypothetical, abstract structure defined by those properties
3. Any claim proven about the abstract structure is true for any concrete model satisfying the same axioms

1.1.6 Groups

Definition 1.1.1 (Group). A group is a set G equipped with an operation $\star$ that satisfies:

- (i) $\forall a, b \in G, a \star b \in G$ (closure)
- (ii) $\forall a, b, c \in G, (a \star b) \star c = a \star (b \star c)$ (associativity)
- (iii) $\exists e \in G$ such that $\forall a \in G, a \star e = e \star a = a$ (neutral element)
- (iv) $\forall a \in G, \exists b \in G$ such that $a \star b = b \star a = e$ (invertibility)

Definition 1.1.2 (Commutative Group). A commutative group is a group that also satisfies $\forall a, b \in G, a \star b = b \star a$.

Example.

- $(\mathbb{Z}, +)$ is a group
- $(\mathbb{Z}, \cdot)$ is not a group (fails invertibility)
- Hours on a clock with "passage of time" operation forms a group

Groups are important because many different things can be modeled as groups, but they don't capture everything we need for linear algebra.

1.1.7 The Rational Numbers Model

Additional observations lead to the need for division. The rational numbers are:

$$\mathbb{Q} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0 \right\}$$

We can represent rational numbers as ordered pairs (a, b) where:

- Equality: $(a, b) = (c, d) \Leftrightarrow ad = bc$
- Addition: $(a, b) + (c, d) = (ad + bc, bd)$
- Multiplication: $(a, b) \cdot (c, d) = (ac, bd)$

1.1.8 Properties of $\mathbb{Q}$

From the definition of operations on $\mathbb{Q}$:

- (i) $\mathbb{Q}$ is closed with respect to both addition and multiplication
- (ii) Both operations are associative
- (iii) Both operations are commutative
- (iv) $(0, 1) \in \mathbb{Q}$ is neutral with respect to addition, $(1, 1) \in \mathbb{Q}$ is neutral with respect to multiplication
- (v) Multiplication is distributive over addition: $a(b + c) = ab + ac$
- (vi) Every $q \in \mathbb{Q}$ is invertible with respect to addition; every $q \neq (0, 1)$ is invertible with respect to multiplication

1.1.9 Fields

Definition 1.1.3 (Field). A field is a set F equipped with two binary operations, called addition and multiplication and denoted $+$ and $\cdot$ respectively, that satisfy:

- (i) $\forall a, b \in F: a + b \in F$ and $a \cdot b \in F$ (closure)
- (ii) $\forall a, b, c \in F: (a + b) + c = a + (b + c)$ and $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ (associativity)
- (iii) $\forall a, b \in F: a + b = b + a$ and $a \cdot b = b \cdot a$ (commutativity)
- (iv) $\forall a, b, c \in F: a \cdot (b + c) = (a \cdot b) + (a \cdot c)$ (distributivity)
- (v) $\exists \tilde{0} \in F$ and $\exists \tilde{1} \in F$ such that $\forall a \in F: a + \tilde{0} = a$ and $a \cdot \tilde{1} = a$ (neutral elements)
- (vi) $\forall a \in F, \exists a' \in F$ such that $a + a' = \tilde{0}$ (additive inverses)
- (vii) $\forall a \neq \tilde{0} \in F, \exists a' \in F$ such that $a \cdot a' = \tilde{1}$ (multiplicative inverses)
- (viii) $\tilde{0} \neq \tilde{1}$ (distinct neutral elements)

Note. The two operations are not completely symmetric

- Multiplication distributes over addition, but not vice versa
- The additive neutral is not required to have a multiplicative inverse

1.1.10 Notation and Terminology

- Elements of a field are called **scalars**
- We drop the tildes and refer to $\tilde{0}$ and $\tilde{1}$ as "zero" and "one"
- Multiplication precedes addition in order of operations
- We often omit the multiplication symbol: $ab + ac$ instead of $a \cdot b + a \cdot c$

1.1.11 Uniqueness of Inverses

Proposition 1.1.4 (Uniqueness of Additive Inverse). Let F be a field and $a \in F$. There exists a unique $a' \in F$ such that $a + a' = 0$.

Proof. Let $a \in F$ and suppose $a', a'' \in F$ such that

$$\begin{aligned} a' + a &= a + a' = 0 \\ a'' + a &= a + a'' = 0 \end{aligned}$$

Then:

$$a' = a' + 0 = a' + (a + a'') = (a' + a) + a'' = 0 + a'' = a''$$

Proposition 1.1.5 (Uniqueness of Multiplicative Inverse). Let F be a field and $a \neq 0 \in F$. There exists a unique $a' \in F$ such that $a \cdot a' = 1$.

Proof. Let $a \in F$ and suppose $a', a'' \in F$ such that

$$\begin{aligned} a' \cdot a &= a \cdot a' = 1 \\ a'' \cdot a &= a \cdot a'' = 1 \end{aligned}$$

Then:

$$a' = a' \cdot 1 = a' \cdot (a \cdot a'') = (a' \cdot a) \cdot a'' = 1 \cdot a'' = a''$$

1.1.12 Inverse Notation and Operations

Since inverses are unique:

- The additive inverse of a is denoted $(-a)$
- The multiplicative inverse of a is denoted a^{-1}
- Subtraction: $a - b := a + (-b)$
- Division: when $b \neq 0, a/b := a \cdot b^{-1}$

1.2 Consequences

Lecture 05: Consequences

1.2.1 Properties of Zero

Proposition 1.2.1 (Zero Property). Let F be a field. Then $\forall a \in F, a \cdot 0 = 0$.

Proof. Let $a \in F$. Then:

$$\begin{aligned} a \cdot \tilde{0} &= a \cdot \tilde{0} + \tilde{0} \\ &= a \cdot \tilde{0} + (a \cdot \tilde{0} + (- (a \cdot \tilde{0}))) = (a \cdot \tilde{0} + a \cdot \tilde{0}) + (- (a \cdot \tilde{0})) \\ &= a \cdot (\tilde{0} + \tilde{0}) + (- (a \cdot \tilde{0})) = a \cdot \tilde{0} + (- (a \cdot \tilde{0})) = \tilde{0} \end{aligned}$$

■

Proposition 1.2.2 (Zero Product Property). Let F be a field. Then $\forall a, b \in F$:

$$a \cdot b = 0 \Rightarrow a = 0 \text{ or } b = 0$$

Proof. Either $a = 0$ or $a \neq 0$.

If $a = 0$, then the conclusion is satisfied.

If $a \neq 0$, then by field axiom (vii), $\exists a^{-1} \in F$ such that $a \cdot a^{-1} = a^{-1} \cdot a = 1$. Therefore:

$$b = b \cdot 1 = b \cdot (a \cdot a^{-1}) = (b \cdot a) \cdot a^{-1} = 0 \cdot a^{-1} = 0$$

Proposition 1.2.3 (Zero Non-Invertible). Let F be a field. Then $0 \in F$ is not invertible with respect to multiplication.

Proof. Suppose, for contradiction, that $\exists a \in F$ such that $a \cdot 0 = 1$.

Then by Proposition 1.2.1, $1 = a \cdot 0 = 0$.

But by field axiom (viii), $0 \neq 1$, which is a contradiction. ■

1.2.2 Modular Arithmetic

The integers modulo m is the set $\mathbb{Z}_m = \{0, 1, 2, \dots, m-1\}$ with addition and multiplication defined by taking remainders after division by m .

Example. $\mathbb{Z}_2 = \{0, 1\}$ with operation tables:

+	0	1
0	0	1
1	1	0

·	0	1
0	0	0
1	0	1

This is a field.

Example. $\mathbb{Z}_3 = \{0, 1, 2\}$ with operation tables:

+	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

·	0	1	2
0	0	0	0
1	0	1	2
2	0	2	1

This is a field.

Example. $\mathbb{Z}_4 = \{0, 1, 2, 3\}$ is not a field because $2 \cdot 2 = 0$ but $2 \neq 0$, violating the zero product property.

Note. These facts are important in applications but not central to linear algebra

- When m is prime, $\mathbb{Z}_m$ is a field
- When m is not prime, $\mathbb{Z}_m$ is not a field (by the zero product property) (doesn't mean no field with m members isn't possible)

1.2.3 Operations Involving Inverses

Proposition 1.2.4 (Inverse Properties). Let F be a field and let $a, b \in F$. Then:

- $-0 = 0$
- $1^{-1} = 1$
- $-(-a) = a$
- $(a^{-1})^{-1} = a$ (when $a \neq 0$)
- $(-1) \cdot a = -a$
- $(-a) \cdot b = -(a \cdot b)$
- $(-a) \cdot b = a \cdot (-b)$
- $(-a) \cdot (-b) = a \cdot b$

Proof.

- Since $0 + 0 = 0$, we have $-0 = 0$ by uniqueness of additive inverses.
- Since $1 \cdot 1 = 1$, we have $1^{-1} = 1$ by uniqueness of multiplicative inverses.
- Since $(-a) + a = 0$, we have $-(-a) = a$ by uniqueness of additive inverses.
- Since $a \cdot a^{-1} \cdot a = 1$, we have $(a^{-1})^{-1} = a$ by uniqueness of multiplicative inverses.
- $a + ((-1) \cdot a) = 1 \cdot a + ((-1) \cdot a) = (1 + (-1)) \cdot a = 0 \cdot a = 0$
- $(-a) \cdot b = ((-1) \cdot a) \cdot b = (-1) \cdot (a \cdot b) = -(a \cdot b)$
- $(-a) \cdot b = ((-1) \cdot a) \cdot b = (-1) \cdot (a \cdot b) = a \cdot ((-1) \cdot b) = a \cdot (-b)$
- $(-a) \cdot (-b) = ((-1) \cdot a) \cdot ((-1) \cdot b) = (-1) \cdot (-1) \cdot a \cdot b$

Since $(-1) \cdot (-1) = (-1)^{-1} \cdot (-1) = 1$, we get: $(-a) \cdot (-b) = 1 \cdot a \cdot b = a \cdot b$ ■

1.3 Numbers

Lecture 06: Numbers

1.3.1 Essential vs Nonessential Properties

Note. The essential properties of natural numbers relate to counting and ordering. Other properties (like primality) are nonessential but useful for examples.

Example. The set $\{0, 1, 2, \dots, p-1\}$ with addition and multiplication modulo p is a field if and only if p is prime.

1.3.2 Number System Extensions

Definition (Integers). Starting with $\mathbb{N}$ and adding zero and additive inverses:

$$\mathbb{Z} = \mathbb{N} \cup \{0\} \cup \{-x : x \in \mathbb{N}\}$$

Definition (Rationals). Taking ordered pairs of integers:

$$\mathbb{Q} = \{(a, b) : a, b \in \mathbb{Z}, b \neq 0\}$$

Note. Algorithms for addition, multiplication, and ordering extend from $\mathbb{N}$ to $\mathbb{Z}$ to $\mathbb{Q}$. The rigorous justification is handled in real analysis.

1.3.3 Irrational Numbers

The Pythagorean theorem reveals distances that cannot be expressed as ratios of integers.

Example (Pythagoras of Samos). Two unit-length sticks at right angles create a distance $\sqrt{2}$ between their free ends. No ratio of integers accurately describes this distance

Note. The existence of irrational numbers has been known since antiquity. Their rigorous construction (e.g., Dedekind cuts) is handled in real analysis.

1.3.4 Real Numbers

Definition (Real Numbers). We accept $\mathbb{R}$ as the union of rational and irrational numbers, using geometric intuition of "length" and the number line.

1.3.5 Complex Numbers

Definition 1.3.1 (Complex Numbers). We denote by $\mathbb{C}$ the set of ordered pairs of real numbers:

$$\mathbb{C} = \{(x, y) : x, y \in \mathbb{R}\}$$

with operations:

$$\text{Equality: } (x, y) = (x', y') \Leftrightarrow x = x' \wedge y = y'$$

$$\text{Addition: } (x, y) + (x', y') = (x + x', y + y')$$

$$\text{Multiplication: } (x, y)(x', y') = (xx' - yy', xy' + x'y)$$

Proposition 1.3.2 (Complex Field). The set $\mathbb{C}$ with addition and multiplication as defined is a field.

Proof. We verify each field axiom:

- Closure:** Since $\mathbb{R}$ is closed under addition and multiplication, both $(x + x', y + y')$ and $(xx' - yy', xy' + x'y)$ belong to $\mathbb{C}$.
- Associativity:** For addition:

$$\begin{aligned} [(x, y) + (x', y')] + (x'', y'') &= (x + x', y + y') + (x'', y'') \\ &= ((x + x') + x'', (y + y') + y'') = (x + (x' + x''), y + (y' + y'')) \\ &= (x, y) + [(x', y') + (x'', y'')] \end{aligned}$$

For multiplication:

$$\begin{aligned} [(x, y)(x', y')](x'', y'') &= (xx' - yy', xy' + x'y)(x'', y'') \\ &= ((xx' - yy')x'' - (xy' + x'y)y'', (xx' - yy')y'' + (xy' + x'y)x'') \\ &= (x(x'x'' - y'y'') - y(x'y'' + y'x''), x(x'y'' + y'x'') + y(x'x'' - y'y'')) \\ &= (x, y)[(x', y')(x'', y'')] \end{aligned}$$

- Commutativity:** For addition:

$$(x, y) + (x', y') = (x + x', y + y') = (x' + x, y' + y) = (x', y') + (x, y)$$

For multiplication:

$$(x, y)(x', y') = (xx' - yy', xy' + x'y) = (x'x - y'y, x'y + y'x) = (x', y')(x, y)$$

- Distributivity:**

$$\begin{aligned} (x, y)[(x', y') + (x'', y'')] &= (x, y)(x' + x'', y' + y'') \\ &= (x(x' + x'') - y(y' + y''), x(y' + y'') + y(x' + x'')) \\ &= (xx' - yy' + xx'' - yy'', xy' + x'y'' + xy'' + yx') \\ &= (x, y)(x', y') + (x, y)(x'', y'') \end{aligned}$$

- Neutral elements:** Let $\vec{0} = (0, 0)$ and $\vec{1} = (1, 0)$. Then:

$$\begin{aligned} (x, y) + (0, 0) &= (x, y) \\ (x, y)(1, 0) &= (x \cdot 1 - y \cdot 0, x \cdot 0 + y \cdot 1) = (x, y) \end{aligned}$$

- Additive inverses:** Let $-(x, y) = (-x, -y)$. Then:

$$(x, y) + (-x, -y) = (0, 0)$$

- Multiplicative inverses:** For $(x, y) \neq (0, 0)$, let:

$$(x, y)^{-1} = \left(\frac{x}{x^2 + y^2}, \frac{-y}{x^2 + y^2} \right)$$

Then $(x, y) \cdot (x, y)^{-1} = (1, 0)$.

- Distinct neutrals:** Since $0 \neq 1$ in $\mathbb{R}$, we have $(0, 0) \neq (1, 0)$. ■

1.3.6 Real Numbers as Complex Subset

The mapping $x \mapsto (x, 0)$ identifies $\mathbb{R}$ with the subset $\{(x, 0) : x \in \mathbb{R}\} \subset \mathbb{C}$.

This preserves operations:

$$x + y \mapsto (x + y, 0) = (x, 0) + (y, 0) \quad (1.1)$$

$$xy \mapsto (xy, 0) = (x, 0)(y, 0) \quad (1.2)$$

1.3.7 The Imaginary Unit

Definition (Imaginary Unit). We define $i = (0, 1)$, which satisfies $i^2 = -1$:

$$i \cdot i = (0, 1)(0, 1) = (0 - 1, 0 + 0) = (-1, 0)$$

Notation. Every complex number can be written uniquely as:

$$(x, y) = (x, 0) + (0, y) = x + yi$$

Definition 1.3.3 (Real and Imaginary Parts). For $z = x + yi \in \mathbb{C}$:

$$\Re(z) = x \quad (\text{real part}) \quad (1.3)$$

$$\Im(z) = y \quad (\text{imaginary part}) \quad (1.4)$$

1.3.8 Geometric Interpretation

Complex numbers correspond to points in the Cartesian plane, with the real part as x -coordinate and imaginary part as y -coordinate.

Polar Form

Definition 1.3.4 (Absolute Value and Argument). For $z = x + yi \in \mathbb{C}$:

$$|z| = \sqrt{x^2 + y^2} \quad (\text{absolute value}) \quad (1.5)$$

$$\arg(z) = \arctan\left(\frac{y}{x}\right) \quad (\text{argument, with correct quadrant}) \quad (1.6)$$

For complex numbers $z = |z|(\cos \theta + i \sin \theta)$ and $z' = |z'|(\cos \theta' + i \sin \theta')$:

$$zz' = |z||z'|[\cos(\theta + \theta') + i \sin(\theta + \theta')]$$

The absolute value of the product equals the product of absolute values, and the argument of the product equals the sum of arguments.

1.3.9 Complex Conjugate

Definition 1.3.5 (Complex Conjugate). For $z = x + yi \in \mathbb{C}$, the complex conjugate is:

$$\bar{z} = x - yi$$

Theorem 1.3.6 (Conjugate Properties). For $z, z_1, z_2 \in \mathbb{C}$:

$$(i) \quad \bar{\bar{z}} = z$$

$$(ii) \quad \overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$$

$$(iii) \quad \overline{z_1 z_2} = \bar{z}_1 \cdot \bar{z}_2$$

$$(iv) \quad z\bar{z} = |z|^2$$

$$(v) \quad z = \bar{z} \Leftrightarrow z \in \mathbb{R}$$

1.3.10 Division

For $z, z' \in \mathbb{C}$ with $z' \neq 0$:

$$\frac{z}{z'} = \frac{z\bar{z'}}{z'z'} = \frac{z\bar{z'}}{|z'|^2} = \frac{xx' + yy'}{x'^2 + y'^2} + i \frac{x'y - y'x}{x'^2 + y'^2}$$

Note. Elementary functions (exponential, trigonometric, logarithmic) extend to $\mathbb{C}$ while preserving many properties. The exponential function leads to what many consider the most beautiful equation in mathematics: Euler's identity, $e^{i\pi} + 1 = 0$, while considering functions brings us to complex analysis.

Chapter 2

Systems of Linear Equations

Lecture 07: A Linear Equation Over a Field

2.1 A Linear Equation Over a Field

Proposition 2.1.1. Let F be a field.

- $\forall a, b, c \in F$
$$a + b = a + c \Rightarrow b = c$$
- $\forall a \neq \tilde{0} \in F$ and $\forall b, c \in F$
$$ab = ac \Rightarrow b = c$$

Proof.

- $\exists (-a) \in F$ such that $(-a) + a = \tilde{0}$. Then
$$a + b = a + c \Rightarrow (-a) + (a + b) = (-a) + (a + c)$$

$$\Rightarrow ((-a) + a) + b = ((-a) + a) + c \Rightarrow \tilde{0} + b = \tilde{0} + c \Rightarrow b = c$$

- $\exists a^{-1} \in F$ such that $a^{-1}a = \tilde{1}$. Then
$$ab = ac \Rightarrow a^{-1}(ab) = a^{-1}(ac)$$

$$\Rightarrow (a^{-1}a)b = (a^{-1}a)c \Rightarrow \tilde{1}b = \tilde{1}c \Rightarrow b = c$$

Definition 2.1.2 (Standard Form Linear Equation). A **Standard form linear Equation in n unknowns over a field F** is an expression of the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

in which the $x_1, \dots, x_n \in F$ are the **unknowns**, the $a_1, \dots, a_n \in F$ are the **coefficients of the unknowns**, and $b \in F$ is the **free coefficient**.

Definition (Linear Equation). An expression that represents the same condition as some standard form linear equation

Note.

- We often drop an explicit 1 or (-1) as a coefficient of an unknown.
- When the number of unknowns is small, it is sometimes convenient to use a, b, c, x, y, z instead.

Definition 2.1.3 (n -tuple). A finite list of n items, not necessarily distinct, appearing in a particular order.

Note. For 2-tuples and 3-tuples, we often use the terms **pair** and **triple** respectively.

Notation.

- Write components of an n -tuple in parentheses, separated by commas.
- We name an n -tuple with roman, lowercase, underlined letters (e.g., $\underline{a} = (a_1, a_2, \dots, a_n)$).

Note. In print, **a** is more common, but harder when writing.

Definition 2.1.4 (F^n). Let F be a field. The set of all n -tuples whose components are elements of F is denoted by F^n :

$$F^n = \{(a_1, a_2, \dots, a_n) : \forall i \in \{1, 2, \dots, n\}, a_i \in F\}$$

Definition 2.1.5 (Solution). Let $a_1x_1 + a_2x_2 + \dots + a_nx_n = b$ be a linear equation over a field F , the n -tuple

$$(c_1, c_2, \dots, c_n)$$

is a **solution** of the equation if this equality is true:

$$a_1c_1 + a_2c_2 + \dots + a_nc_n = b$$

To “solve” an equation with n unknowns over a field F means to characterize precisely the subset of F^n of solutions of that equation.

Example. Let us solve this equation over $\mathbb{Q}$

$$2x - 3y = 7$$

Using the uniqueness of additive and multiplicative inverses in $\mathbb{Q}$, we deduce

$$2x - 3y = 7 \Leftrightarrow -3y = 7 - 2x \Leftrightarrow 3y = 2x - 7 \Leftrightarrow y = \frac{2}{3}x - \frac{7}{3}$$

Therefore, for any $t \in \mathbb{Q}$, the 2-tuple $(t, \frac{2}{3}t - \frac{7}{3})$ is a solution.

We can express the set of solutions, S in set builder notation:

$$S = \left\{ \left(t, \frac{2}{3}t - \frac{7}{3} \right) : t \in \mathbb{Q} \right\}$$

Proposition 2.1.6. Consider the linear equation $a_1x_1 + a_2x_2 + \dots + a_nx_n = b$ over a field F .

- If $a_i = \tilde{0} = b, \forall i \in \{1, \dots, n\}$ then every n -tuple in F^n is a solution. The solution set is F^n .
- If $a_i = \tilde{0}, \forall i \in \{1, \dots, n\}$ and $b \neq \tilde{0}$ then no n -tuple in F^n is a solution. The solution set is $\emptyset$.
- If at least one of the $a_i \neq \tilde{0}$ then there exists a solution.

Proof.

- By Proposition 1.2.1, for any $(x_1, \dots, x_n) \in F^n$,

$$\tilde{0}x_1 + \tilde{0}x_2 + \dots + \tilde{0}x_n = \tilde{0} = b$$

- By Proposition 1.2.1, for any $(x_1, \dots, x_n) \in F^n$,

$$\tilde{0}x_1 + \tilde{0}x_2 + \dots + \tilde{0}x_n = \tilde{0} \neq b$$

- If $a_i \neq \tilde{0}$ then a_i^{-1} exists and, for any $(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \in F^{n-1}$, the n -tuple

$$(x_1, \dots, x_{i-1}, a_i^{-1}(b - a_1x_1 - \dots - a_{i-1}x_{i-1} - a_{i+1}x_{i+1} - \dots - a_nx_n), x_{i+1}, \dots, x_n)$$

is a solution. ■

2.2 A System of Linear Equations

Lecture 08: A System of Linear Equations

2.2.1 System Definition

Definition 2.2.1 (Standard Form Linear System). A **standard form linear system of order $m \times n$** over a field F is a collection of standard form linear equations:

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$\vdots$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

where the a_{ij} and b_i (for $1 \leq i \leq m, 1 \leq j \leq n$) are scalars of F , called the **coefficients of the system**. The a_{ij} are the **coefficients of the unknowns** and the b_i are the **free coefficients**.

Definition 2.2.2 (Solution of a System). An n -tuple $\underline{c} = (c_1, \dots, c_n) \in F^n$ is a **solution** of an $m \times n$ linear system if it is a solution of each of the m equations of the system.

2.2.2 Consistency

The solution set of a linear system is the intersection of the individual solution sets of each equation.

Definition 2.2.3 (Consistency). If the solution set of a system of linear equations is empty, the system is called **inconsistent**. If the solution set contains at least one solution, the system is called **consistent**.

Note. If any equation in the system has the form:

$$0x_1 + 0x_2 + \dots + 0x_n = b \neq 0$$

then the system is inconsistent. Later, we'll call this a “type-B row”

Note. The algorithm for solving linear systems will transform any inconsistent system to one that is obviously inconsistent.

2.2.3 Redundancy

Definition 2.2.4 (Redundancy). A linear system is called **redundant** if one or more of the equations can be dropped without changing the solution set.

Note. The algorithm for solving linear systems will transform any redundant system to one that is obviously redundant.

2.2.4 Homogeneous Systems

Definition 2.2.5 (Homogeneous System). A linear system is called **homogeneous** if all its free coefficients are zeros. If at least one free coefficient is non-zero, the system is called **non-homogeneous**.

Proposition 2.2.6 (Homogeneous Consistency). Every homogeneous system is consistent.

Proof. Every homogeneous system has at least one solution, namely the n -tuple of zeros: $\underline{0} = (0, 0, \dots, 0)$ ■

Definition 2.2.7 (Trivial Solution). The solution $\underline{0} = (0, 0, \dots, 0)$ is called the **trivial solution** of a homogeneous system.

Note. A homogeneous system may or may not have other solutions besides the trivial solution.

There are no non-homogenous systems where $(\vec{0}, \vec{0}, \vec{0})$ is a solution.

Lecture 09: Equivalent Systems

2.3 Equivalent Systems

Definition 2.3.1 (Equivalent). Two linear systems are called **equivalent** if their solution sets are identical.

Definition 2.3.2 (Elementary Change). An **elementary change** to a linear system is one of the follow changes:

1. Swapping the position of two equations
2. Multiplying every coefficient in one equation by a non-zero scalar
3. Adding the same multiple of the coefficients of one equation to the respective coefficients of another equation

Proposition 2.3.3. Performing an elementary change on a linear system does not change the system's solution set.

Proof. Type (i) Let S be the solution set of a linear system and let S' be the solution set of the system following the elementary change $R_i \leftrightarrow R_j$.

The system includes the same equations before and after the change, so clearly $S = S'$. ■

Proof. Type (ii)

Let S be the solution set of a linear system and let S' be the solution set of the system following the elementary change $R_i \rightarrow tR_i$, where $t \neq 0 \in F$.

All equations except the i -th remain unchanged. For the i -th equation:

Suppose the n -tuple $c = (c_1, c_2, \dots, c_n) \in S$, then

$$a_{i1}c_1 + a_{i2}c_2 + \dots + a_{in}c_n = b_i$$

Therefore,

$$ta_{i1}c_1 + ta_{i2}c_2 + \dots + ta_{in}c_n = t(a_{i1}c_1 + a_{i2}c_2 + \dots + a_{in}c_n) = tb_i$$

Therefore c is also a solution of the i -th equation in the changed system, and $c \in S'$.

Conversely, suppose $c \in S'$, then

$$ta_{i1}c_1 + ta_{i2}c_2 + \dots + ta_{in}c_n = tb_i$$

Therefore, because $t \neq 0$,

$$\begin{aligned} a_{i1}c_1 + \dots + a_{in}c_n &= t^{-1}(ta_{i1}c_1 + ta_{i2}c_2 + \dots + ta_{in}c_n) \\ &= t^{-1}(tb_i) = b_i \end{aligned}$$

Therefore, c is also a solution of the i -th equation in the original system, and $c \in S$. Since $S \subseteq S'$ and $S' \subseteq S$, therefore $S = S'$. ■

Proof. Type (iii)

Let S be the solution set of a linear system and let S' be the solution set of the system following the elementary change $R_i \rightarrow R_i + tR_j$ for $t \in F$.

All equations except the i -th remain unchanged. For the i -th equation:

Suppose the n -tuple $c = (c_1, \dots, c_n) \in S$ is a solution of the original system, then

$$a_{i1}c_1 + a_{i2}c_2 + \dots + a_{in}c_n = b_i$$

and

$$a_{j1}c_1 + a_{j2}c_2 + \dots + a_{jn}c_n = b_j$$

Therefore,

$$\begin{aligned} (ta_{j1} + a_{i1})c_1 + \dots + (ta_{jn} + a_{in})c_n \\ &= ta_{j1}c_1 + \dots + ta_{jn}c_n + a_{i1}c_1 + \dots + a_{in}c_n \\ &= t(a_{j1}c_1 + \dots + a_{jn}c_n) + (a_{i1}c_1 + \dots + a_{in}c_n) \\ &= tb_j + b_i \end{aligned}$$

This shows that c is also a solution of the i -th equation in the changed system and therefore $c \in S'$ and $S \subseteq S'$.

Conversely, suppose $c \in S'$ is a solution of the changed system, then

$$(a_{i1} + ta_{j1})c_1 + (a_{i2} + ta_{j2})c_2 + \dots + (a_{in} + ta_{jn})c_n = b_i + tb_j$$

and

$$a_{j1}c_1 + a_{j2}c_2 + \dots + a_{jn}c_n = b_j$$

Therefore,

$$\begin{aligned} a_{i1}c_1 + \dots + a_{in}c_n \\ &= (a_{i1}c_1 + \dots + a_{in}c_n) + 0 \\ &= (a_{i1}c_1 + \dots + a_{in}c_n) - ((a_{i1} + ta_{j1})c_1 + \dots + (a_{in} + ta_{jn})c_n) + (b_i + tb_j) \\ &= (a_{i1}c_1 + \dots + a_{in}c_n) - (a_{i1}c_1 + \dots + a_{in}c_n) + b_i - t(a_{j1}c_1 + \dots + a_{jn}c_n) + \\ &= 0 + b_i - tb_j + tb_j \\ &= b_i \end{aligned}$$

This shows that c is also a solution of the i -th equation in the original system and therefore $c \in S$ and $S' \subseteq S$.

Therefore, since $S' \subseteq S$ and $S \subseteq S'$, we have $S = S'$. ■

Theorem 2.3.4. A **finite sequence of elementary changes** on a linear system leads to an **equivalent** system.

Proof. Let S be the solution set of a linear system and S' be the solution set of the system after performing a finite sequence of elementary changes.

By Proposition 2.3.3, any single elementary change leads to an equivalent system.

Equivalency is transitive, because equality of sets is transitive. ■

Lecture 10: Gaussian Elimination

2.4 Gaussian Elimination with Back Substitution

2.4.1 Coefficient Matrix

Definition 2.4.1 (Matrix). A **matrix** is a rectangular table of scalars from a field.

The number of rows and columns is called the **order** of the matrix. A matrix of order $m \times n$ matrix has m rows and n columns. Individual scalar values are called the **entries** of the matrix.

Notation. We usually name matrices with uppercase Roman letters and matrix elements have the corresponding lowercase letter. For example:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Definition (Augmented Coefficient Matrix). Any linear system of m equations with n unknowns is represented fully by an $m \times (n + 1)$ matrix, called the **augmented coefficient matrix**:

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{bmatrix}$$

Definition 2.4.2 (Row Equivalent). Let A and B be matrices of the same order. A is **row equivalent** to B if we can change A into B by a finite sequence of elementary row operations.

Note. Row equivalence is a valid equivalence relation because it is reflexive, symmetric, and transitive.

Definition 2.4.3 (Pivot). A matrix row containing all zeros is called a **null row**. In any row that is not a null row, the leftmost non-zero entry is called the **leading coefficient** or the **pivot**.

2.4.2 Row Echelon Form

Definition 2.4.4 (REF). A matrix is said to be in **row echelon form** (REF) if:

1. In every row that is not a null row the leading coefficient is to the right of all the leading coefficients of the rows above it (if any)
2. All null rows, if any exist, are below all non-null rows

Definition 2.4.5 (eliminated system). A linear system whose augmented coefficient matrix is in row echelon form

Definition 2.4.6 (Restricted and Free Variables). In an eliminated linear system, x_j is called a **restricted variable** if there is a pivot in the j -th column of its coefficient matrix. Otherwise, it is called a **free variable**.

Note. In a solution set of an eliminated system, the restricted variables are specified in terms of the free variables. Each free variable in the general solution n -tuple is represented by a free parameter.

2.4.3 Gaussian Elimination

Given a matrix order of $m \times n$

1. If all entries are zero, the matrix is already in REF. Stop
2. If the matrix has only one row, it is already in REF. Stop
3. Identify the first (leftmost) column containing a non-zero entry, column j
4. If the first (topmost) entry in that column is zero, swap with column with non-zero entry in column j
5. If there are any non-zero entries left in column j below the topmost row, add multiples of the topmost row to the rows of those entries to make them zero
6. Ignore the topmost row and repeat steps 1 – 5 on the remaining sub-matrix

Note. This algorithm works on any field

Lecture 11: Gauss-Jordan Elimination

2.5 Gauss-Jordan Elimination

Definition 2.5.1 (RREF). A matrix is in **reduced row echelon form** (RREF) if it is in REF and:

1. All leading coefficients are $\vec{1}$
2. In a column that includes a leading coefficient, all other coefficients are $\vec{0}$

Note. This definition applies in every field

Given a matrix of order $m \times n$

1. Consider row $i = 1$
2. Identify the first (leftmost) column containing a non-zero entry on or below row i , column j
3. If entry $a_{ij} = 0$, swap row i with any row $k > i$ that doesn't have a zero in column j

- Now that $a_{ij} \neq 0$, multiply row i by a_{ij}^{-1}
- If there are any non-zero entries left in column j , add multiples of row i to the rows of those entries to make them zero.
- Increase $i \rightarrow i + 1$ and repeat steps 2 – 5

However, for solving a system of linear equations, RREF is not worth it (REF is faster and easy enough ($\frac{1}{2}O(n^3)$ vs $\frac{1}{3}O(n^3) + \frac{1}{2}O(n^2)$))

Lecture 12: How Many Solutions Are There?

2.6 System Solution Analysis

2.6.1 Consistency

Theorem 2.6.1 (Consistency Condition). A linear system is consistent if and only if its augmented coefficient matrix is row equivalent to a matrix in reduced row echelon form that does not contain a type-B row.

Proof. Form the augmented coefficient matrix of the system and run the Gauss-Jordan elimination algorithm.

- All operations during the algorithm are elementary row operations
- All matrices (initial, final, and intermediate) are row equivalent to the original matrix
- All linear systems represented by these matrices are equivalent
- If at any point a type-B row appears, the system is inconsistent
- If the algorithm completes, the output is a RREF matrix without a type-B row
- This matrix represents a system equivalent to the original system
- A solution is available by back substitution

2.6.2 Number of Solutions

Let A be a linear system with n unknowns over a field F , and suppose the augmented coefficient matrix is row equivalent to a RREF that does not contain a type-B row.

- If the RREF matrix has exactly n non-zero rows, then the system has a **unique solution**
- If the RREF matrix has $k < n$ non-zero rows, then there are $n - k$ free variables:
 - If field F has infinitely many scalars, there are **infinitely many solutions**
 - If field F has N different scalars, there are N^{n-k} solutions

Note. The RREF matrix has at most n non-zero rows because each non-zero row has a pivot in one of the first n columns, and everything else in a pivot column is zero.

2.6.3 Homogeneous Systems

Note. A homogeneous system is always consistent because the trivial solution always exists.

Theorem 2.6.2 (Non-trivial Solutions). A homogeneous system with more unknowns than equations has a non-trivial solution.

Proof. Let A be a homogeneous system with m equations in n unknowns where $m < n$.

- The RREF of the coefficient matrix has no type-B rows (system is homogeneous)
- There are at most m pivots in the RREF matrix (at most one per row)
- Therefore, there are at most m restricted variables
- Therefore, there is at least one free variable since $n - m > 0$
- Any choice of $t \in F$ for the free variable leads to a solution that is an n -tuple with t as one component
- Including the choice $t = 1$
- Since $1 \neq 0$ by axiom viii in Definition 1.1.3, there exists a non-trivial solution

Lecture 13: Square Systems and Square Matrices

2.7 Square Matrices

An $n \times n$ matrix is called a **square matrix**. A linear system with n equations in n unknowns has a square coefficient matrix.

Notation. To avoid confusion with coefficient matrices, we will use the following terminology:

- Augmented coefficient matrix:** includes the constants column $[A|b]$
- Not-augmented coefficient matrix:** contains only the coefficients of the unknowns

In a square matrix of order n , the **main diagonal** is the diagonal from upper left to lower right, consisting of entries a_{ii} for $i \in \{1, 2, \dots, n\}$.

Definition 2.7.1 (Identity Matrix). The square matrix of order n whose main diagonal entries are equal to $\bar{1}$ and all other entries are equal to $\bar{0}$ is called the **identity matrix of order n** , denoted I_n .

$$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

The element in row i , column j of I_n equals δ_{ij} (Kronecker delta).

2.7.1 RREF and Identity Matrix

Theorem 2.7.2 (RREF Square Matrix). If A is an RREF square matrix of order n with a pivot in every row, then $A = I_n$.

Proof. Let A be as described. Since there's a pivot in every row, there are n pivots total.

In RREF, every pivot is in a different column. Suppose the pivot of the i -th row is in the j -th column.

- There are $i - 1$ rows above the i -th row, whose pivots must be in columns strictly to the left of column j , so $i - 1 \leq j - 1$
- There are $n - i$ rows below the i -th row, whose pivots must be in columns strictly to the right of column j , so $n - i \leq n - j$

Therefore $i \leq j$ and $j \leq i$, which implies $i = j$.

The pivots of A are therefore a_{ii} for $i \in \{1, \dots, n\}$. In RREF:

- All pivots equal $\bar{1}$
- In every pivot column, all non-pivot entries equal $\bar{0}$

Therefore $a_{ij} = \delta_{ij}$, which is the definition of I_n . ■

2.7.2 Unique Solutions for Square Systems

Lemma 2.7.3. If A and B are row equivalent, and we delete column j from each to make A' and B' respectively, then A' and B' are also row equivalent.

Proof. The same sequence of row operations that leads from A to B also leads from A' to B' . ■

Theorem 2.7.4 (Unique Solution Test). A linear system of n equations in n unknowns has a unique solution if and only if its not-augmented coefficient matrix is row equivalent to I_n .

Proof. Let $A = [A'|b_\uparrow]$ be the augmented coefficient matrix, where A' is the not-augmented coefficient matrix.

($\Rightarrow$): Suppose A' is row equivalent to I_n .

Apply the same sequence of operations to $A = [A'|b_\uparrow]$ making $[I_n|c_\uparrow]$

This represents the system: $x_1 = c_1, x_2 = c_2, \dots, x_n = c_n$

This system has the unique solution $\underline{c} = (c_1, c_2, \dots, c_n)$

($\Leftarrow$): Suppose the system has a unique solution.

Apply row operations to get RREF matrix $C = [C'|c_\uparrow]$

Since the original system has one solution, so does the equivalent RREF system

Therefore, there are no free variables, so C has n pivots in the first n columns

Therefore C' has a pivot in every row and every column

By Theorem 2.7.2, $C' = I_n$

By Lemma 2.7.3, A' is row equivalent to I_n ■

Corollary 2.7.5. For a square matrix A : if one linear system with coefficient matrix A has a unique solution, then **all** linear systems with coefficient matrix A have unique solutions.

Equivalently: A square system $[A|b]$ has a unique solution if and only if the corresponding homogeneous system $[A|0]$ has no non-trivial solution.

2.7.3 Square Homogeneous Systems

Theorem 2.7.6 (Square Homogeneous Dichotomy). For the not-augmented coefficient matrix of a homogeneous system of n equations in n unknowns, exactly one of the following is true:

- It is row equivalent to a matrix with a row of zeros $\Leftrightarrow$ the system has a non-trivial solution
- It is row equivalent to $I_n \Leftrightarrow$ the system has no non-trivial solution

Proof. Proof of (1):

- ($\Rightarrow$) Suppose A is row equivalent to a matrix B with a row of zeros
- WLOG, assume the zero row is the last row (swap if necessary)
- Apply row operations to get RREF matrix C (the zero row remains)
- A square RREF matrix with a zero row has fewer than n pivots
- Therefore the system has at least one free variable
- Therefore the homogeneous system has a non-trivial solution
- ($\Leftarrow$) If A has a non-trivial solution, it doesn't have a unique solution
- By Theorem 2.7.4, A is not row equivalent to I_n
- By Theorem 2.7.2, the RREF of A doesn't have a pivot in every row
- Therefore it has a row of zeros

Proof of (2):

- ($\Rightarrow$) If A is row equivalent to I_n , the system has a unique solution
- Since it's homogeneous, that unique solution is the trivial solution
- ($\Leftarrow$) If the system has no non-trivial solution, by part (1) it's not row equivalent to a matrix with zero rows
- Therefore the RREF of A has no zero rows
- By Theorem 2.7.2, the RREF of A is I_n ■

Chapter 3

Matrix Algebra

Lecture 14: Definition and Notation

1 Notation

Definition 3.1.1 (Matrix). A **matrix** is a rectangular table of scalars from a field.

- The number of rows and columns is called the **order** of the matrix
- A matrix of order $m \times n$ (read: "em-by-en") has m rows and n columns
- The individual scalar values are called the **entries** of the matrix

In general, a matrix A of order $m \times n$ is written as:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Notation.

- To emphasize the matrix order, we may write $A_{m \times n}$
- The entry in row i , column j of A is called the (i, j) entry and denoted a_{ij}
- Rarely, a comma is needed to distinguish, e.g., $a_{11,2}$ from $a_{1,12}$
- A matrix A of order $m \times n$ with entries a_{ij} is denoted $A = [a_{ij}]_{m \times n}$
- An anonymous matrix of order $m \times n$ with entries a_{ij} is denoted $[a_{ij}]_{m \times n}$
- An anonymous (i, j) entry of A is denoted $[A]_{ij}$

Definition 3.1.2 (Row and Column Vectors).

- A matrix of order $1 \times n$ is called a **row matrix** or **row vector** of order n
- A matrix of order $n \times 1$ is called a **column matrix** or **column vector** of order n

Notation.

- The i -th row of matrix A is denoted $[A]_i^r$
- The j -th column of matrix A is denoted $[A]_j^c$

Example. If $A = [a_{ij}]_{m \times n}$, then for all $i \in \{1, \dots, m\}$ and $j \in \{1, \dots, n\}$:

$$[A]_i^r = [a_{i1} \quad \cdots \quad a_{in}]$$

$$[A]_j^c = \begin{bmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{bmatrix}$$

Definition 3.1.3 (Transpose). Let $A = [a_{ij}]_{m \times n}$. The **transpose** of A , denoted A^t , is the $n \times m$ matrix whose (i, j) entry is the (j, i) entry of A :

$$[A^t]_{ij} = a_{ji}$$

Definition 3.1.4 (Square Matrix). A matrix of order $n \times n$ is called a **square matrix** of order n .

Definition 3.1.5 (Diagonals). Let $A = [a_{ij}]_{n \times n}$ be a square matrix of order n .

- The n -tuple $(a_{11}, a_{22}, \dots, a_{nn})$ is called the **main diagonal** of A
- The n -tuple $(a_{1n}, a_{2, n-1}, \dots, a_{n1})$ is called the **secondary diagonal** of A

Definition 3.1.6 (Diagonal Matrix). A matrix $[a_{ij}]_{n \times n}$ is called a **diagonal matrix** if $i \neq j \Rightarrow a_{ij} = 0$.

Definition 3.1.7 (Matrix Equality). The matrices $A_{m \times n}$ and $B_{p \times q}$ are **equal** if $m = p$ and $n = q$ and for every $i \in \{1, \dots, m\}$ and $j \in \{1, \dots, n\}$, $[A]_{ij} = [B]_{ij}$.

Definition 3.1.8 (Symmetric Matrix). A matrix A is called a **symmetric matrix** if $A^t = A$.

Notation. To refer to the set of all matrices of order $m \times n$ with entries from the field F , we write

$$M_{m \times n}(F)$$

For square matrices of order n , we write simply

$$M_n(F)$$

Lecture 15: Operations with Matrices

3.2 Operations

Definition 3.2.1 (Matrix Addition). Let $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{m \times n}$ be matrices with entries in the same field F . The **sum** $A + B$ is the matrix

$$A + B := [a_{ij} + b_{ij}]_{m \times n}$$

Note. Matrix addition is only defined for matrices of the same order. For example, the sum

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 6 & 7 \\ 8 & 9 & 0 \end{bmatrix}$$

is undefined.

Proposition 3.2.2 (Properties of Matrix Addition). Matrix addition satisfies the following properties:

- Closure:** For all $A, B \in M_{m \times n}(F)$, $A + B \in M_{m \times n}(F)$
- Commutativity:** For all $A, B \in M_{m \times n}(F)$, $A + B = B + A$
- Associativity:** For all $A, B, C \in M_{m \times n}(F)$, $(A + B) + C = A + (B + C)$
- Existence of neutral element:** There exists $O \in M_{m \times n}(F)$ such that for all $A \in M_{m \times n}(F)$, $A + O = A$
- Existence of additive inverse:** For every $A \in M_{m \times n}(F)$, there exists $B \in M_{m \times n}(F)$ such that $A + B = O$

Proof. Let $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{m \times n}$ be matrices in $M_{m \times n}(F)$.

(i) **Closure:** By definition, $A + B = [a_{ij} + b_{ij}]_{m \times n}$. Since $a_{ij}, b_{ij} \in F$ and F is closed under addition, $a_{ij} + b_{ij} \in F$. Therefore $A + B \in M_{m \times n}(F)$.

(ii) **Commutativity:** For all i, j , $[A + B]_{ij} = a_{ij} + b_{ij} = b_{ij} + a_{ij} = [B + A]_{ij}$ by commutativity of addition in F . Therefore $A + B = B + A$.

(iii) **Associativity:** For all i, j , $[(A + B) + C]_{ij} = (a_{ij} + b_{ij}) + c_{ij} = a_{ij} + (b_{ij} + c_{ij}) = [A + (B + C)]_{ij}$ by associativity of addition in F . Therefore $(A + B) + C = A + (B + C)$.

(iv) **Neutral element:** Let $O = [0]_{m \times n}$ be the matrix whose entries are all zeros of F . Then for all i, j , $[A + O]_{ij} = a_{ij} + 0 = a_{ij} = [A]_{ij}$. Therefore $A + O = A$.

(v) **Additive inverse:** Let $B = [-a_{ij}]_{m \times n}$, where $-a_{ij}$ is the additive inverse of a_{ij} in F . Then for all i, j , $[A + B]_{ij} = a_{ij} + (-a_{ij}) = 0 = [O]_{ij}$. Therefore $A + B = O$. ■

Note. The additive neutral element is called the **zero matrix**. As usual, associativity implies generalized associativity, so we can omit parentheses in sums.

Definition 3.2.3 (Matrix Subtraction). Let $A, B \in M_{m \times n}(F)$. The **difference** is defined as

$$A - B := A + (-B)$$

where $-B$ is the additive inverse of B .

Definition 3.2.4 (Scalar Multiplication). Let $A = [a_{ij}]_{m \times n} \in M_{m \times n}(F)$ and let $t \in F$. The **scalar multiple** tA is the matrix

$$tA := [ta_{ij}]_{m \times n}$$

Theorem 3.2.5 (Properties of Scalar Multiplication). For any $A, B \in M_{m \times n}(F)$ and any $t, s \in F$:

- $tA \in M_{m \times n}(F)$
- $s(tA) = (st)A$
- $(s + t)A = sA + tA$
- $t(A + B) = tA + tB$
- $1A = A$
- $0A = O_{m \times n}$
- $(-1)A = -A$

Definition 3.2.6 (Matrix Multiplication). Let $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{n \times q}$ be matrices with entries from the same field F . The **product** AB is the $m \times q$ matrix defined by

$$[AB]_{ij} = \sum_{k=1}^n a_{ik} b_{kj} = a_{i1} b_{1j} + a_{i2} b_{2j} + \cdots + a_{in} b_{nj}$$

Note. Key observations about matrix multiplication:

- The product AB is defined when the number of columns in A equals the number of rows in B
- The resulting matrix AB has as many rows as A and as many columns as B
- To compute $[AB]_{ij}$, multiply each entry in row i of A by the corresponding entry in column j of B , then sum these n products
- If A is a $1 \times n$ row matrix and B is an $n \times 1$ column matrix, then AB is a 1×1 matrix (scalar)
- If A is an $n \times 1$ column matrix and B is a $1 \times n$ row matrix, then BA is an $n \times n$ square matrix
- If A is not square, then AA is not defined

Note. Matrix multiplication is defined this way because it corresponds to composition of linear transformations. If F is the matrix representing a linear function f and G represents g , then FG represents the composition $f \circ g$, but only when this product is defined as in Definition 3.2.6.

Lecture 16: Properties of Matrix Multiplication

3.3 Properties

Proposition 3.3.1 (Zero Matrix Multiplication). (i) For all $A \in M_{m \times n}(F)$ and all $p \in \mathbb{N}$, $A_{m \times n} O_{n \times p} = O_{m \times p}$
(ii) For all $B \in M_{n \times p}(F)$ and all $m \in \mathbb{N}$, $O_{m \times n} B_{n \times p} = O_{m \times p}$

Proof. (i): Let $A = [a_{ij}]_{m \times n}$. Then for any $p \in \mathbb{N}$ and for all i, j ,

$$[A_{m \times n} O_{n \times p}]_{ij} = \sum_{k=1}^n [A]_{ik} [O]_{kj} = \sum_{k=1}^n a_{ik} \cdot 0 = \sum_{k=1}^n 0 = 0$$

(ii): Let $B = [b_{ij}]_{n \times p}$. Then for any $m \in \mathbb{N}$ and for all i, j ,

$$[O_{m \times n} B_{n \times p}]_{ij} = \sum_{k=1}^n [O]_{ik} [B]_{kj} = \sum_{k=1}^n 0 \cdot b_{kj} = \sum_{k=1}^n 0 = 0$$

Lemma 3.3.2 (Row and Column Products). Let $A = [a_{ij}]_{m \times n}$, $B = [b_{ij}]_{n \times p}$, and let $C = [c_{ij}]_{m \times p} = AB$. Then:

- (i) $[C]_i^r = [A]_i^r B$
- (ii) $[C]_j^c = A [B]_j^c$
- (iii) $[c_{ij}]_{1 \times 1} = [A]_i^r [B]_j^c$

Proof. (i): The LHS is a $1 \times p$ matrix whose $(1, j)$ entry is c_{ij} . The RHS is a product of a $1 \times n$ matrix with an $n \times p$ matrix, resulting in a $1 \times p$ matrix whose $(1, j)$ entry is $\sum_{k=1}^n a_{ik} b_{kj} = c_{ij}$.

(ii): The LHS is an $m \times 1$ matrix whose $(i, 1)$ entry is c_{ij} . The RHS is a product of an $m \times n$ matrix with an $n \times 1$ matrix, resulting in an $m \times 1$ matrix whose $(i, 1)$ entry is $\sum_{k=1}^n a_{ik} b_{kj} = c_{ij}$.

(iii): Both the LHS and RHS are 1×1 matrices with the single element $c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$. ■

Proposition 3.3.3 (Zero Row/Column Preservation). Let $A \in M_{m \times n}(F)$ and $B \in M_{n \times p}(F)$.

- (i) If the i -th row of A is all zeros, then the i -th row of AB is all zeros
- (ii) If the j -th column of B is all zeros, then the j -th column of AB is all zeros

Proof. This follows directly from Lemma 3.3.2 and Proposition 3.3.1. ■

Proposition 3.3.4 (Transpose of Product). Let $A \in M_{m \times n}(F)$ and $B \in M_{n \times p}(F)$. Then

$$(AB)^t = B^t A^t$$

Proof. First, we verify the orders match. Since AB is $m \times p$, we have $(AB)^t$ is $p \times m$. Also, B^t is $p \times n$ and A^t is $n \times m$, so $B^t A^t$ is defined and is $p \times m$.

Now we verify the entries are equal. For all i, j :

$$\begin{aligned} [(AB)^t]_{ij} &= [AB]_{ji} = \sum_{k=1}^n a_{jk} b_{ki} = \sum_{k=1}^n b_{ki} a_{jk} = \sum_{k=1}^n [B]_{ki} [A]_{jk} \\ &= \sum_{k=1}^n [B^t]_{ik} [A^t]_{kj} = [B^t A^t]_{ij} \end{aligned}$$

Theorem 3.3.5 (Associativity of Matrix Multiplication). Let $A = [a_{ij}]_{m \times n}$, $B = [b_{ij}]_{n \times p}$, and $C = [c_{ij}]_{p \times q}$ be matrices with entries in the same field F . Then

$$(AB)C = A(BC)$$

Proof. First, we verify the orders. AB is defined and is $m \times p$, so $(AB)C$ is defined and is $m \times q$. Also, BC is defined and is $n \times q$, so $A(BC)$ is defined and is $m \times q$.

Now we verify the entries are equal. For all i, j :

$$\begin{aligned} [(AB)C]_{ij} &= \sum_{k=1}^p [AB]_{ik} c_{kj} = \sum_{k=1}^p \left(\sum_{l=1}^n a_{il} b_{lk} \right) c_{kj} = \sum_{k=1}^p \sum_{l=1}^n a_{il} b_{lk} c_{kj} \\ &= \sum_{l=1}^n \sum_{k=1}^p a_{il} b_{lk} c_{kj} = \sum_{l=1}^n a_{il} \left(\sum_{k=1}^p b_{lk} c_{kj} \right) = \sum_{l=1}^n a_{il} [BC]_{lj} \\ &= [A(BC)]_{ij} \end{aligned}$$

Theorem 3.3.6 (Distributivity of Matrix Multiplication). Let $A = [a_{ij}]_{m \times n} \in M_{m \times n}(F)$ and $B = [b_{ij}]_{n \times n} \in M_{n \times n}(F)$.

- (i) **Right distributivity:** For any $C = [c_{ij}]_{n \times p} \in M_{n \times p}(F)$,

$$(A + B)C = AC + BC$$

- (ii) **Left distributivity:** For any $D = [d_{ij}]_{q \times m} \in M_{q \times m}(F)$,

$$D(A + B) = DA + DB$$

Proof. (i): For all i, j ,

$$\begin{aligned} [(A + B)C]_{ij} &= \sum_{k=1}^n [A + B]_{ik} c_{kj} = \sum_{k=1}^n (a_{ik} + b_{ik}) c_{kj} \\ &= \sum_{k=1}^n (a_{ik} c_{kj} + b_{ik} c_{kj}) = \sum_{k=1}^n a_{ik} c_{kj} + \sum_{k=1}^n b_{ik} c_{kj} \\ &= [AC]_{ij} + [BC]_{ij} = [AC + BC]_{ij} \end{aligned}$$

Proposition 3.3.7 (Scalar Factoring). Let $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{n \times p}$, and let $t \in F$. Then:

- (i) $t(AB) = (tA)B$
- (ii) $t(AB) = A(tB)$

Proof. (i): For all i, j ,

$$\begin{aligned} [t(AB)]_{ij} &= t[AB]_{ij} = t \sum_{k=1}^n a_{ik} b_{kj} = \sum_{k=1}^n t a_{ik} b_{kj} \\ &= \sum_{k=1}^n [tA]_{ik} b_{kj} = [(tA)B]_{ij} \end{aligned}$$

Definition 3.3.8 (Identity Matrix). The **identity matrix of order n** , denoted I_n , is the matrix

$$I_n = [\delta_{ij}]_{n \times n}$$

where δ_{ij} is the Kronecker delta ($\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$). In other words, I_n is the square matrix of order n whose main diagonal entries are all equal to 1 and all other entries are equal to zero.

Theorem 3.3.9 (Identity Matrix Property). For all $A = [a_{ij}]_{m \times n}$:

- (i) $I_m A = A$
- (ii) $A I_n = A$

Proof. (i): The shapes work out: I_m is $m \times m$ and A is $m \times n$, so $I_m A$ is $m \times n$. For all i, j ,

$$\begin{aligned} [I_m A]_{ij} &= \sum_{k=1}^m [I_m]_{ik} [A]_{kj} = \sum_{k=1}^m \delta_{ik} [A]_{kj} \\ &= 0 \cdot [A]_{1j} + 0 \cdot [A]_{2j} + \cdots + 1 \cdot [A]_{ij} + \cdots + 0 \cdot [A]_{mj} = [A]_{ij} \end{aligned}$$

Lecture 17: Fun with Squares

3.4 Squares

Note. The set $M_n(F)$ has special properties with respect to matrix multiplication:

- $M_n(F)$ is closed with respect to matrix multiplication (the product of two $n \times n$ matrices is always defined and is also $n \times n$)
- The matrix I_n is the neutral element with respect to multiplication in $M_n(F)$
- The neutral element with respect to a binary operation is always unique; I_n is the only matrix satisfying $I_n A = A I_n = A$ for all $A \in M_n(F)$

Note. However, this does not mean we cannot find $A, B \in M_n(F)$ with $I_n \neq A, B$ such that $AB = BA = A$.

Note. In general, for matrices $A \in M_{m \times n}(F)$ and $B \in M_{p \times q}(F)$:

- AB is not always defined
- Even when AB is defined, BA is not always defined
- Even when AB and BA are both defined, they might not have the same shape
- Even when AB and BA are both defined and have the same shape, they might not be equal

Definition 3.4.1 (Commuting Matrices). Two square matrices A and B of the same order are said to **commute** if

$$AB = BA$$

Note. A necessary condition for $AB = BA$ is that A and B must both be square and of the same order.

Definition 3.4.2 (Scalar Matrix). A matrix $A \in M_n(F)$ is called a **scalar matrix** if there exists $t \in F$ such that $A = tI_n$. That is, a scalar matrix is a diagonal matrix where all diagonal entries are equal.

Result 3.4.3 (Scalar Matrices Commute). Any scalar matrix commutes with any square matrix of the same order.

Proof. Let $A, B \in M_n(F)$ and suppose A is scalar. Then there exists $t \in F$ such that $A = tI_n$. Therefore:

$$\begin{aligned} AB &= (tI_n)B \\ &= t(I_n B) && \text{(Proposition 3.3.7(i))} \\ &= tB && \text{(Theorem 3.3.9(i))} \\ &= t(BI_n) && \text{(Theorem 3.3.9(ii))} \\ &= (tI_n)B && \text{(Proposition 3.3.7(ii))} \\ &= B(tI_n) && \text{(Proposition 3.3.7(i))} \\ &= BA \end{aligned}$$

■

Theorem 3.4.4 (Diagonal Matrices Commute). Diagonal matrices of the same order commute.

Proof. Let $A, B \in M_n(F)$ be diagonal. Then AB and BA are both defined and are square of order n . For all $i, j \in \{1, \dots, n\}$:

For AB :

$$[AB]_{ij} = \sum_{k=1}^n [A]_{ik}[B]_{kj}$$

Since A is diagonal, $[A]_{ik} = 0$ whenever $i \neq k$. The only nonzero term in the sum occurs when $k = i$:

$$[AB]_{ij} = [A]_{ii}[B]_{ij}$$

Since B is diagonal, $[B]_{ij} = 0$ when $i \neq j$. Therefore:

$$[AB]_{ij} = [A]_{ii}[B]_{ii}\delta_{ij}$$

Similarly, for BA :

$$[BA]_{ij} = \sum_{k=1}^n [B]_{ik}[A]_{kj}$$

Since B is diagonal, the only nonzero term occurs when $k = i$:

$$[BA]_{ij} = [B]_{ii}[A]_{ij}$$

Since A is diagonal, $[A]_{ij} = 0$ when $i \neq j$. Therefore:

$$[BA]_{ij} = [B]_{ii}[A]_{ii}\delta_{ij} = [A]_{ii}[B]_{ii}\delta_{ij}$$

Thus $[AB]_{ij} = [BA]_{ij}$ for all i, j , so $AB = BA$. ■

Note. We have found sufficient conditions for $AB = BA$:

- At least one of them is scalar
- Both of them are diagonal

However, there is no simple necessary and sufficient condition for commutativity.

3.4.2 Matrix Powers

Definition 3.4.5 (Matrix Power). Let $A \in M_n(F)$ be a square matrix and let $p \in \mathbb{N} \cup \{0\}$. The **matrix power** A^p is defined recursively:

- If $p = 0$: $A^0 := I_n$
- If $p > 0$: $A^p := A^{p-1}A$

Note. This definition gives us:

- $A^1 = A$
- $A^2 = AA$
- $A^3 = AAA$
- And so on

Proposition 3.4.6 (Powers Commute). If B and C are powers of the same matrix A , then they commute.

Proof. Let $B = A^r$ and $C = A^s$ for some $r, s \in \mathbb{N} \cup \{0\}$. By associativity of matrix multiplication:

$$A^r A^s = \underbrace{A \cdot A \cdots A}_{r+s \text{ times}} = A^s A^r$$

Therefore $BC = CB$. ■

Example. Consider the Fibonacci sequence defined by:

- $a_1 = 1$
- $a_2 = 1$
- $a_n = a_{n-1} + a_{n-2}$ for all $n > 2$

Matrix powers can be used to compute a_{32} without computing all of a_n for $n \in \{1, \dots, 31\}$. (This technique uses the matrix $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ and its powers.)

Lecture 18: Matrix Equations

3.5 Equations

Note. Now that we know how to perform operations on matrices and check equality, we can write equations with matrices. For example:

- $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} X = \begin{bmatrix} 4 & 5 & 6 \end{bmatrix}$
- $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} X = \begin{bmatrix} 7 & 8 & 9 \end{bmatrix}$
- $A + tX^3 = B$

Note. Important differences from scalar equations:

- Not every matrix equation has a solution
- Unlike scalar algebraic equations, the various terms in a matrix equation are not necessarily from the same set (they may have different dimensions)

3.5.1 The Matrix Equation $AX = B$

Note. Suppose $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{m \times 1}$. What is the shape of a matrix that may be a solution to $AX = B$?

For the product AX to be defined and have shape $m \times 1$, we need $X = [x_{ij}]_{n \times 1}$. Recall that a matrix of order $n \times 1$ is called a column vector (Definition 3.1.2).

Notation. For a column vector $X_{n \times 1}$, we also use the notation $\mathbf{x}$ (or simply x when context is clear). There is a natural one-to-one correspondence between F^n (the set of n -tuples) and $M_{n \times 1}(F)$ (the set of column vectors).

3.5.2 Connection to Linear Systems

Let $A = [a_{ij}]_{m \times n}$ and $\mathbf{b} = [b_i]_{m \times 1}$. Consider the matrix equation

$$A\mathbf{x} = \mathbf{b}$$

Expanding using the definitions of matrix equality and matrix multiplication, we have:

$$A\mathbf{x} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

This gives us:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

Theorem 3.5.1 (Matrix Form of Linear Systems). The matrix equation $A\mathbf{x} = \mathbf{b}$ is equivalent to a system of m linear equations in n unknowns, where A is the coefficient matrix and $\mathbf{b}$ is the constants column vector.

Note. This compact notation provides a powerful way to represent and work with linear systems. We will use this formulation when we return to solving systems of linear equations.

Lecture 19: Regular and Singular Matrices

3.6 Regulars

Note. The matrix equation $A\mathbf{x} = \mathbf{b}$ is equivalent to a linear system. A column vector $\mathbf{x}$ is a solution of the matrix equation if and only if the n -tuple x is a solution of the linear system.

3.6.1 Solving Matrix Equations

Note. Writing $A\mathbf{x} = \mathbf{b}$ instead of the full linear system is not merely convenient notation. It allows us to manipulate equations using matrix algebra. Can we "solve" the system in one step as $\mathbf{x} = \mathbf{b}/A$?

Note. Consider the scalar equation $ax = b$ where $a, b \in F$. We can solve for x as $x = b/a$ only if $a \neq 0$. When $a \neq 0$, we have $x = a^{-1}b$. The key property: in a field, every nonzero element has a multiplicative inverse.

Note. However, $M_{m \times n}(F)$ is not a field. Even $M_n(F)$ is not a field, though it has better properties:

- Multiplication is always defined and closed: $A, B \in M_n(F) \Rightarrow AB \in M_n(F)$
- There is a multiplicative neutral element: $I_n \in M_n(F)$ such that $AI_n = I_n A = A$ for all $A \in M_n(F)$
- Where there is a neutral element, there may be inverses

Note. Not all matrices have inverses:

- The zero matrix O_n has no inverse (similar to how 0 has no inverse in a field)
- Any matrix with a row of zeros has no inverse (if A has a zero row, then AB has a zero row, so $AB \neq I$)
- Any matrix with a column of zeros has no inverse (if A has a zero column, then BA has a zero column, so $BA \neq I$)
- $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ has no inverse (verify by attempting to find one)

Definition 3.6.1 (Regular and Singular Matrices). A matrix $A \in M_n(F)$ is called **regular** (or **invertible**) if there exists $B \in M_n(F)$ such that

$$AB = BA = I_n$$

In this case, B is called an **inverse** of A . A matrix that is not regular is called **singular** (or **non-invertible**).

Notation. When an inverse of A exists, it is unique. We call it **the inverse** and denote it A^{-1} .

Note. Only square matrices can be regular or singular. While you may find $A \in M_{2 \times 3}(F)$ and $B \in M_{3 \times 2}(F)$ such that $AB = I_2$ and $BA = I_3$, these are different identity matrices, so this doesn't constitute invertibility.

Theorem 3.6.2 (One-Sided Inverse Test). Let $A, B \in M_n(F)$.

- (i) If A is regular and $AB = I$, then $BA = I$
- (ii) If A is regular and $BA = I$, then $AB = I$
- (iii) If $AB = I_n$, then A is regular and B is its inverse (proof deferred to Unit 5)

In words: if you know that A is regular and suspect that B is its inverse, you only need to verify one of the two products equals I .

Proof. (i): If A is regular, then A^{-1} exists and

$$BA = I(BA) = (A^{-1}A)(BA) = A^{-1}(AB)A = A^{-1}(I)A = A^{-1}A = I$$

(ii): If A is regular, then A^{-1} exists and

$$AB = (AB)I = (AB)(AA^{-1}) = A(BA)A^{-1} = A(I)A^{-1} = AA^{-1} = I$$

■

Proposition 3.6.3 (Cancellation Laws). Let $A \in M_n(F)$ be regular. Then for any $B, C \in M_n(F)$:

- (i) $AB = AC \Rightarrow B = C$
- (ii) $BA = CA \Rightarrow B = C$

Proof. (i): If $AB = AC$, multiply both sides on the left by A^{-1} :

$$A^{-1}(AB) = A^{-1}(AC) \Rightarrow (A^{-1}A)B = (A^{-1}A)C \Rightarrow IB = IC \Rightarrow B = C$$

(ii): If $BA = CA$, multiply both sides on the right by A^{-1} :

$$(BA)A^{-1} = (CA)A^{-1} \Rightarrow B(AA^{-1}) = C(AA^{-1}) \Rightarrow BI = CI \Rightarrow B = C$$

■

Proposition 3.6.4 (Zero Product Properties). Let $A, B \in M_n(F)$.

- (i) If A is regular and $AB = O_n$, then $B = O_n$
- (ii) If A, B are both regular, then $AB \neq O_n$
- (iii) If $A \neq O_n$ and $B \neq O_n$ and $AB = O_n$, then A, B are both singular

Proof. (i): If $AB = O_n$, then $AB = AO_n$. By Proposition 3.6.3(i), $B = O_n$.
(ii): If both A and B are regular and $AB = O_n$, then by (i), $B = O_n$. But O_n is not regular, contradiction.
(iii): Suppose A is regular. Then by (i), $B = O_n$, contradicting $B \neq O_n$. Similarly, if B is regular, then $A = O_n$. Therefore both must be singular. ■

Theorem 3.6.5 (Properties of Inverses). Let $A, B \in M_n(F)$.

- (i) If A is regular, then A^{-1} is also regular and $(A^{-1})^{-1} = A$
- (ii) If A is regular, then A^t is also regular and $(A^t)^{-1} = (A^{-1})^t$
- (iii) If A, B are both regular, then AB is also regular and $(AB)^{-1} = B^{-1}A^{-1}$
- (iv) If A is regular and $s \neq 0 \in F$, then sA is also regular and $(sA)^{-1} = s^{-1}A^{-1}$

Proof. (i): We have $A^{-1}A = AA^{-1} = I$. This shows A is an inverse of A^{-1} , so A^{-1} is regular with $(A^{-1})^{-1} = A$.

(ii): We need to show $(A^{-1})^t A^t = A^t (A^{-1})^t = I$. Using Proposition 3.3.4:

$$(A^{-1})^t A^t = (AA^{-1})^t = I^t = I$$

$$A^t (A^{-1})^t = (A^{-1}A)^t = I^t = I$$

(iii): We need to show $(AB)(B^{-1}A^{-1}) = (B^{-1}A^{-1})(AB) = I$:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$$

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I$$

(iv): We need to show $(sA)(s^{-1}A^{-1}) = (s^{-1}A^{-1})(sA) = I$:

$$(sA)(s^{-1}A^{-1}) = s(A(s^{-1}A^{-1})) = s(s^{-1}(AA^{-1})) = s(s^{-1}I) = (ss^{-1})I = I$$

$$(s^{-1}A^{-1})(sA) = s^{-1}(A^{-1}(sA)) = s^{-1}(s(A^{-1}A)) = s^{-1}(sI) = (s^{-1}s)I = I$$

■

Theorem 3.6.6 (Product of Regular Matrices). For all $k \in \mathbb{N}$, if $A_1, A_2, \dots, A_k \in M_n(F)$ are regular, then $A_1 A_2 \dots A_k$ is regular and

$$(A_1 A_2 \dots A_{k-1} A_k)^{-1} = A_k^{-1} A_{k-1}^{-1} \dots A_2^{-1} A_1^{-1}$$

Proof. By induction on k .

Base case ($k = 2$): This is Theorem 3.6.5(iii).

Induction hypothesis: Suppose for some $k \in \mathbb{N}$, if $A_1, \dots, A_k \in M_n(F)$ are regular, then $A_1 \dots A_k$ is regular and $(A_1 \dots A_k)^{-1} = A_k^{-1} \dots A_1^{-1}$.

Induction step: Let $A_1, \dots, A_k, A_{k+1} \in M_n(F)$ be regular. Let $B = A_1 A_2 \dots A_k$. By the induction hypothesis, B is regular and $B^{-1} = A_k^{-1} \dots A_1^{-1}$. By Theorem 3.6.5(iii), BA_{k+1} is regular and

$$(BA_{k+1})^{-1} = A_{k+1}^{-1} B^{-1} = A_{k+1}^{-1} A_k^{-1} \dots A_1^{-1}$$

Therefore $A_1 \dots A_k A_{k+1}$ is regular with the claimed inverse. ■

Note. Regular matrices have properties analogous to "nonzero" numbers. For example, you can move a regular matrix to the other side of an equation, or cancel it if it appears on both sides. But how do we determine if a given matrix is regular? And if it is regular, how do we find its inverse?

Definition 3.7.1 (Elementary Matrix). $A \in M_n(F)$ is called **elementary** if we can get A by performing one elementary row operation on I_n .

Notation. If we denote an elementary row operation by ϕ , then $\phi(A)$ denotes the matrix obtained by performing the operation ϕ on the matrix A . Any elementary matrix is therefore $\phi(I_n)$ for some elementary row operation ϕ .

Proposition 3.7.2 (Elementary Matrix Multiplication). Let $A \in M_n(F)$ and let ϕ be an elementary row operation. Then

$$\phi(A) = \phi(I_n)A$$

Proof. We use Lemma 3.3.2: $[AB]_k^r = [A]_k^r B$. We consider each type of elementary row operation.

Case 1: Let $\phi : R_i \leftrightarrow R_j$.

For any $k \neq i, j$:

$$[\phi(A)]_k^r = [A]_k^r \quad \text{and} \quad [\phi(I)]_k^r = [I]_k^r$$

Therefore:

$$[\phi(I)A]_k^r = [\phi(I)]_k^r A = [I]_k^r A = [IA]_k^r = [A]_k^r = [\phi(A)]_k^r$$

For $k = i$:

$$[\phi(A)]_i^r = [A]_j^r \quad \text{and} \quad [\phi(I)]_i^r = [I]_j^r$$

Therefore:

$$[\phi(I)A]_i^r = [\phi(I)]_i^r A = [I]_j^r A = [IA]_j^r = [A]_j^r = [\phi(A)]_i^r$$

Similarly for $k = j$. Therefore $[\phi(I)A]_k^r = [\phi(A)]_k^r$ for all k .

Case 2: Let $\phi : R_i \rightarrow tR_i$.

For $k \neq i$:

$$[\phi(A)]_k^r = [A]_k^r \quad \text{and} \quad [\phi(I)]_k^r = [I]_k^r$$

Therefore:

$$[\phi(I)A]_k^r = [\phi(I)]_k^r A = [I]_k^r A = [IA]_k^r = [A]_k^r = [\phi(A)]_k^r$$

For $k = i$:

$$[\phi(A)]_i^r = t[A]_i^r \quad \text{and} \quad [\phi(I)]_i^r = t[I]_i^r$$

Therefore:

$$[\phi(I)A]_i^r = [\phi(I)]_i^r A = t[I]_i^r A = t([I]_i^r A) = t[IA]_i^r = t[A]_i^r = [\phi(A)]_i^r$$

Case 3: Let $\phi : R_i \rightarrow R_i + tR_j$.

For $k \neq i$:

$$[\phi(A)]_k^r = [A]_k^r \quad \text{and} \quad [\phi(I)]_k^r = [I]_k^r$$

Therefore:

$$[\phi(I)A]_k^r = [A]_k^r = [\phi(A)]_k^r$$

For $k = i$:

$$[\phi(A)]_i^r = [A]_i^r + t[A]_j^r \quad \text{and} \quad [\phi(I)]_i^r = [I]_i^r + t[I]_j^r$$

Therefore:

$$[\phi(I)A]_i^r = [\phi(I)]_i^r A = ([I]_i^r + t[I]_j^r)A = [I]_i^r A + t[I]_j^r A = [A]_i^r + t[A]_j^r = [\phi(A)]_i^r$$

In all cases, $[\phi(I)A]_k^r = [\phi(A)]_k^r$ for all k , so $\phi(A) = \phi(I)A$. ■

Corollary 3.7.3 (Sequence of Elementary Operations). Suppose B is the matrix obtained by performing the elementary operations $\phi_1, \phi_2, \dots, \phi_{k-1}, \phi_k$ (in that order) on the matrix A . Then

$$B = \phi_k(I)\phi_{k-1}(I)\dots\phi_2(I)\phi_1(I)A$$

Proof. By induction on k .

Base case ($k = 1$): This is Proposition 3.7.2.

Induction hypothesis: If C is the matrix obtained by performing $\phi_1, \phi_2, \dots, \phi_k$ on A , then

$$C = \phi_k(I)\dots\phi_1(I)A$$

Induction step: Suppose B is obtained by performing $\phi_1, \phi_2, \dots, \phi_k, \phi_{k+1}$ on A . Then B is obtained by performing the single operation ϕ_{k+1} on C . Thus:

$$B = \phi_{k+1}(C) = \phi_{k+1}(I)C = \phi_{k+1}(I)\phi_k(I)\dots\phi_1(I)A$$

■

Conclusion 3.7.4 (Reverse Operation). Let ϕ be an elementary row operation. The **reverse operation**, denoted ϕ^{-1} , is defined as:

- If $\phi : R_i \leftrightarrow R_j$, then $\phi^{-1} : R_i \leftrightarrow R_j$ (note: $\phi = \phi^{-1}$)
- If $\phi : R_i \rightarrow tR_i$, then $\phi^{-1} : R_i \rightarrow t^{-1}R_i$ (note: $t \neq 0 \in F$)
- If $\phi : R_i \rightarrow R_i + tR_j$, then $\phi^{-1} : R_i \rightarrow R_i - tR_j$ (note: $i \neq j$)

For any ϕ and A : $\phi^{-1}(\phi(A)) = A$ and $(\phi^{-1})^{-1} = \phi$.

Conclusion 3.7.5 (Elementary Matrices are Regular). Every elementary matrix $\phi(I)$ is regular, and its inverse is $\phi^{-1}(I)$.

Lecture 20: Elementary Matrices

3.7 Elementary

Proof. For any A and any ϕ :

$$\phi^{-1}(\phi(A)) = A = \phi(\phi^{-1}(A))$$

By Corollary 3.7.3:

$$\phi^{-1}(I)\phi(I)A = A = \phi(I)\phi^{-1}(I)A$$

This holds for any A , including $A = I$:

$$\phi^{-1}(I)\phi(I)I = I = \phi(I)\phi^{-1}(I)I$$

Therefore:

$$\phi^{-1}(I)\phi(I) = I \quad \text{and} \quad \phi(I)\phi^{-1}(I) = I$$

So $\phi(I)$ has an inverse, namely $\phi^{-1}(I)$, which means $[\phi(I)]^{-1} = \phi^{-1}(I)$. ■

Proposition 3.7.6 (Products of Elementary Matrices are Regular). Let $A \in M_n(F)$. If

$$A = \phi_1(I)\phi_2(I) \cdots \phi_{k-1}(I)\phi_k(I)$$

then A is regular, and its inverse is

$$A^{-1} = \phi_k^{-1}(I)\phi_{k-1}^{-1}(I) \cdots \phi_2^{-1}(I)\phi_1^{-1}(I)$$

In words: any matrix that is a product of finitely many elementary matrices is regular.

Proof. Each $\phi_i(I)$ is regular by Proposition ??, with $[\phi_i(I)]^{-1} = \phi_i^{-1}(I)$. By Theorem 3.6.6, the product of finitely many regular matrices is regular, and:

$$[\phi_1(I) \cdots \phi_k(I)]^{-1} = [\phi_k(I)]^{-1} \cdots [\phi_1(I)]^{-1} = \phi_k^{-1}(I) \cdots \phi_1^{-1}(I)$$

■

Theorem 3.7.7 (Characterization of Regular Matrices). If $A \in M_n(F)$ is regular, then A is a product of elementary matrices.

Proof. Let $A \in M_n(F)$ be regular. Consider the matrix equation

$$A\mathbf{x} = \mathbf{0}$$

Multiplying both sides from the left by A^{-1} :

$$A^{-1}(A\mathbf{x}) = A^{-1}\mathbf{0} \Rightarrow (A^{-1}A)\mathbf{x} = \mathbf{0} \Rightarrow I\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x} = \mathbf{0}$$

Therefore this equation has only one solution, namely $\mathbf{x} = \mathbf{0}$.

Since A is the coefficient matrix of a homogeneous system of n equations in n unknowns with only the trivial solution, by Theorem 2.7.6, A is row-equivalent to I_n .

Therefore, there exists a sequence of elementary row operations $\phi_1, \dots, \phi_k$ such that:

$$\phi_k(\cdots \phi_1(A) \cdots) = I_n$$

By Corollary 3.7.3, is a sequence of elementary matrices $\phi_1(I_n), \dots, \phi_k(I_n)$ such that:

$$\phi_k(I_n) \cdots \phi_1(I_n)A = BA = I_n$$

where $B = \phi_k(I_n) \cdots \phi_1(I_n)$.

By Proposition 3.7.6, B is regular with inverse:

$$B^{-1} = \phi_1^{-1}(I_n) \cdots \phi_k^{-1}(I_n)$$

Since $BA = I_n$, we have $A = B^{-1}$. Therefore:

$$A = \phi_1^{-1}(I_n) \cdots \phi_k^{-1}(I_n)$$

which is a product of elementary matrices. ■

Proof. (i): If A and B are both regular, then by Theorem 3.8.1, each is row equivalent to I_n , and therefore they are row equivalent to each other.

(ii): If A and B are row equivalent and one is row equivalent to I_n , then the other is also row equivalent to I_n . By Theorem 3.8.1, either both are regular or both are singular.

(iii): Since A and B are row equivalent, there exists a sequence of elementary row operations relating them. By Corollary 3.7.3, $B = CA$ where C is a product of elementary matrices. By Proposition 3.7.6, C is regular. ■

Note. Given $A \in M_n(F)$, we perform Gauss-Jordan elimination while keeping track of the row operations. There are exactly two possibilities:

- (i) We end with a matrix that has a row of zeros
- (ii) We end with the identity matrix I_n

Note. Case (i): If we end with a row of zeros, then A is not regular.

Case (ii): If we end with I_n , then A is regular. The Gauss-Jordan elimination found that:

$$\phi_k(\cdots \phi_1(A) \cdots) = I_n$$

Therefore:

$$\phi_k(I_n) \cdots \phi_1(I_n)A = I_n$$

Since A is regular, we have:

$$A^{-1} = \phi_k(I_n) \cdots \phi_1(I_n) = \phi_k(I_n) \cdots \phi_1(I_n)I_n = \phi_k(\cdots \phi_1(I_n) \cdots)$$

In words: We construct the inverse of A by performing on I_n the same row operations, in the same order, that were performed on A during Gauss-Jordan elimination.

Theorem 3.8.3 (Characterizations of Regularity). Let $A \in M_n(F)$. The following are equivalent (each is true if and only if any of the others are true):

- (i) A is regular
- (ii) A is a product of elementary matrices
- (iii) A is row equivalent to I_n
- (iv) There exists a regular matrix C such that $CA = I$
- (v) For every column vector $\mathbf{b} \in M_{n \times 1}(F)$, there is a unique solution to the equation $A\mathbf{x} = \mathbf{b}$
- (vi) The equation $A\mathbf{x} = \mathbf{0}$ does not have a nontrivial solution

Proof. Most of these are restatements of previous results:

(i) $\Leftrightarrow$ (ii): This follows from Proposition 3.7.6 and Theorem 3.7.7.

(i) $\Leftrightarrow$ (iii): This is Theorem 3.8.1.

(iii) $\Leftrightarrow$ (iv): This is a particular case of Proposition 3.8.2(iii) with $B = I$.

(i) $\Rightarrow$ (v): If A is regular, then $\mathbf{x} = A^{-1}\mathbf{b}$ is the unique solution to $A\mathbf{x} = \mathbf{b}$.

(v) $\Rightarrow$ (vi): If (v) holds, then for $\mathbf{b} = \mathbf{0}$, there is a unique solution. Since $\mathbf{x} = \mathbf{0}$ is a solution, it must be the unique solution, so there is no nontrivial solution.

(vi) $\Rightarrow$ (iii): This is a restatement of Theorem 2.7.6(ii). ■

Lecture 21: How to Invert a Matrix

3.8 Inverting

Theorem 3.8.1 (Regularity and Row Equivalence). A matrix $A \in M_n(F)$ is regular if and only if it is row equivalent to I_n .

Proof. ($\Rightarrow$): Suppose A is row equivalent to I_n . Then there exists a sequence of elementary row operations $\phi_1, \dots, \phi_k$ such that:

$$A = \phi_1(\phi_2(\cdots \phi_k(I_n) \cdots))$$

Therefore:

$$A = \phi_1(I_n)\phi_2(I_n) \cdots \phi_k(I_n)I_n = \phi_1(I_n)\phi_2(I_n) \cdots \phi_k(I_n)$$

So A is a product of elementary matrices, and therefore A is regular by Proposition 3.7.6.

($\Leftarrow$): Suppose A is regular. By Theorem 3.7.7, A is a product of elementary matrices. Therefore there exists a sequence of elementary matrices $\phi_1(I), \dots, \phi_k(I)$ such that:

$$A = \phi_1(I_n)\phi_2(I_n) \cdots \phi_k(I_n)$$

Therefore:

$$A = \phi_1(I_n)\phi_2(I_n) \cdots \phi_k(I_n)I_n = \phi_1(\phi_2(\cdots \phi_k(I_n) \cdots))$$

So A is row equivalent to I_n . ■

Proposition 3.8.2 (Properties of Row Equivalence). (i) Any two regular matrices are row equivalent to each other

(ii) Two row equivalent matrices are either both regular or both singular

(iii) If A, B are row equivalent (regular or singular), there exists a regular C such that $B = CA$

Chapter 4

Vector Spaces

Lecture 22: Arrows

4.1 Vector Arrows

Definition 4.1.1 (Plane Vector). A **plane vector** is a directed line segment in the plane (an arrow). The point of origin is called its **tail**, and the end point is called its **head**.

Note. While vectors have a tail and a head, their most important characteristics are their length and the direction they point in.

Definition 4.1.2 (Same Direction). Two vectors are in the **same direction** if:

- They lie on a single line and, if we translate them along said line until their tails coincide, their heads lie on the same ray of this line that originates from this point; or
- They lie on parallel lines and, if their heads and tails are connected by straight line segments, a quadrilateral is formed.

Definition 4.1.3 (Opposite Directions). Two vectors are in **opposite directions** if they lie on the same line or on parallel lines and they are not in the same direction.

Definition 4.1.4 (Equal Vectors). Two vectors that have the same length and are in the same direction are **equal**. Equal vectors whose tails and heads do not coincide are sometimes referred to as **copies** of the same vector. Any point in the plane is a copy of the vector of zero length.

Definition 4.1.5 (Vector Addition). We add a plane vector to another plane vector by choosing a copy of the second vector whose tail coincides with the head of the first vector and then drawing an arrow from the tail of the first to the head of the (copy of the) second.

Notation. Addition of plane vectors $\vec{a}$ and $\vec{b}$ is denoted $\vec{a} \oplus \vec{b}$.

Proposition 4.1.6 (Commutativity of Vector Addition). For any two plane vectors $\vec{a}, \vec{b}$:

$$\vec{a} \oplus \vec{b} = \vec{b} \oplus \vec{a}$$

Proof. We verify this by considering all possible cases of vector orientations:

- If $\vec{a}$ and $\vec{b}$ are in the same direction, then both $\vec{a} \oplus \vec{b}$ and $\vec{b} \oplus \vec{a}$ form the diagonal of a degenerate parallelogram (a line segment).
- If $\vec{a}$ and $\vec{b}$ are in opposite directions (whether of same length or not), both sums form the same resulting vector.
- If $\vec{a}$ and $\vec{b}$ are neither in the same nor opposite directions (forming either an acute or obtuse angle), then by the parallelogram law, both $\vec{a} \oplus \vec{b}$ and $\vec{b} \oplus \vec{a}$ represent the diagonal of the same parallelogram.

In all cases, the two sums yield equal vectors. ■

Proposition 4.1.7 (Associativity of Vector Addition). For any three plane vectors $\vec{a}, \vec{b}, \vec{c}$:

$$(\vec{a} \oplus \vec{b}) \oplus \vec{c} = \vec{a} \oplus (\vec{b} \oplus \vec{c})$$

Proof. Consider three vectors $\vec{a}, \vec{b}, \vec{c}$.

For $(\vec{a} \oplus \vec{b}) \oplus \vec{c}$: First form $\vec{a} \oplus \vec{b}$ by placing the tail of $\vec{b}$ at the head of $\vec{a}$, then place the tail of $\vec{c}$ at the head of this sum. The result is a vector from the tail of $\vec{a}$ to the head of $\vec{c}$.

For $\vec{a} \oplus (\vec{b} \oplus \vec{c})$: First form $\vec{b} \oplus \vec{c}$ by placing the tail of $\vec{c}$ at the head of $\vec{b}$, then place the tail of this sum at the head of $\vec{a}$. The result is again a vector from the tail of $\vec{a}$ to the head of $\vec{c}$.

Both computations yield the same vector from the tail of $\vec{a}$ to the head of $\vec{c}$ (following the chain $\vec{a} \rightarrow \vec{b} \rightarrow \vec{c}$). ■

Note. As with any associative binary operation, we can drop the parentheses and write, for any finite number of arrows:

$$\vec{a}_1 \oplus \vec{a}_2 \oplus \cdots \oplus \vec{a}_n$$

with no parentheses and no ambiguity.

Note. As with any binary operation that is both commutative and associative, in a finite sum the terms can be written in any order, and the sequence of binary additions carried out in any order. For example:

$$\vec{a} \oplus \vec{b} \oplus \vec{c} \oplus \vec{d} = (\vec{a} \oplus \vec{b}) \oplus (\vec{c} \oplus \vec{d}) = \vec{a} \oplus (\vec{c} \oplus (\vec{b} \oplus \vec{d}))$$

Note. Analogous definitions of vectors and vector addition work the same way in three-dimensional space; geometry in space is harder to visualize.

Lecture 23: Descartes

4.2 Cartesian Coordinates

Note. A Cartesian reference frame consists of:

- A point in the plane chosen as the origin
- Two perpendicular directions chosen as the x and y axes

This framework establishes a one-to-one-to-one correspondence between:

- 2-tuples $(a_1, a_2) \in \mathbb{R}^2$
- Points $\mathbf{a}$ in the plane with x -coordinate a_1 and y -coordinate a_2
- Plane vectors $\vec{a}$ with tail at the origin and head at $\mathbf{a}$

We refer to members of $\mathbb{R}^2$ as 2-tuples, points, or vectors almost interchangeably, as context dictates.

Definition 4.2.1 (Equality of n -Tuples). The n -tuple $a = (a_1, a_2, \dots, a_n)$ and the m -tuple $b = (b_1, b_2, \dots, b_m)$ are said to be **equal**, denoted by $a = b$, if $n = m$ and $a_i = b_i$ for all i , $1 \leq i \leq n$.

Definition 4.2.2 (Addition of n -Tuples). Let F be a field. The **sum** of $a = (a_1, \dots, a_n)$ and $b = (b_1, \dots, b_n)$ in F^n is denoted $a + b$ and defined by

$$a + b := (a_1 + b_1, \dots, a_n + b_n)$$

Definition 4.2.3 (Scalar Multiplication). Let F be a field. **Multiplication** of $a = (a_1, \dots, a_n) \in F^n$ by a scalar $t \in F$ is denoted by the juxtaposition ta and defined by

$$ta := (ta_1, \dots, ta_n)$$

Theorem 4.2.4 (Properties of Addition on F^n). Let F be a field and $n \in \mathbb{N}$. Then:

- The set F^n is closed with respect to n -tuple addition: $\forall a, b \in F^n, (a + b) \in F^n$.
- n -tuple addition is associative: $\forall a, b, c \in F^n, (a + b) + c = a + (b + c)$.
- n -tuple addition is commutative: $\forall a, b \in F^n, a + b = b + a$.
- The n -tuple $0 = (\vec{0}, \dots, \vec{0})$ (where $\vec{0} \in F$ is the additive identity in F) is the additive identity in F^n : $\forall a \in F^n, a + 0 = a$.
- All elements of F^n are invertible with respect to n -tuple addition. For every $a = (a_1, \dots, a_n) \in F^n$, the n -tuple $-a = (-a_1, \dots, -a_n)$ is the unique element of F^n satisfying $a + (-a) = 0$.

Proof. Each property follows immediately from the definition of n -tuple addition and the corresponding properties of addition in F . ■

Notation. As usual, the uniqueness of the additive inverse allows a shorthand notation called subtraction:

$$a - b := a + (-b)$$

Theorem 4.2.5 (Properties of Scalar Multiplication). Let F be a field and $n \in \mathbb{N}$. For every $a, b \in F^n$ and every $t, s \in F$:

- $\vec{0}a = 0$ (where $\vec{0}$ is the additive identity in F and 0 is the additive identity in F^n)
- $\vec{1}a = a$ (where $\vec{1}$ is the multiplicative identity in F)
- $(-\vec{1})a = -a$
- $s(ta) = (st)a$
- $(s + t)a = sa + ta$
- $t(a + b) = ta + tb$

Proof. We prove each property by showing equality of corresponding components.

Property i: For $a = (a_1, \dots, a_n)$, we have $\vec{0}a = (\vec{0}a_1, \dots, \vec{0}a_n) = (\vec{0}, \dots, \vec{0}) = 0$ since $\vec{0}a_i = \vec{0}$ in F for all i .

Property ii: $\vec{1}a = (\vec{1}a_1, \dots, \vec{1}a_n) = (a_1, \dots, a_n) = a$ since $\vec{1}a_i = a_i$ in F for all i .

Property iii: $(-\vec{1})a = ((-\vec{1})a_1, \dots, (-\vec{1})a_n) = (-a_1, \dots, -a_n) = -a$ since $(-\vec{1})a_i = -a_i$ in F for all i .

Property iv: $s(ta) = s(ta_1, \dots, ta_n) = (s(ta_1), \dots, s(ta_n)) = ((st)a_1, \dots, (st)a_n) = (st)a$ by associativity of multiplication in F .

Property v: $(s + t)a = ((s + t)a_1, \dots, (s + t)a_n) = (sa_1 + ta_1, \dots, sa_n + ta_n) = (sa_1, \dots, sa_n) + (ta_1, \dots, ta_n) = sa + ta$ by distributivity in F .

Property vi: For $a = (a_1, \dots, a_n)$ and $b = (b_1, \dots, b_n)$, we have $t(a + b) = t(a_1 + b_1, \dots, a_n + b_n) = (t(a_1 + b_1), \dots, t(a_n + b_n)) = (ta_1 + tb_1, \dots, ta_n + tb_n) = (ta_1, \dots, ta_n) + (tb_1, \dots, tb_n) = ta + tb$ by distributivity in F . ■

Note. Thanks to property iii, we can use the usual shorthand notation: $ta - sb := ta + (-s)b$.

Lemma 4.2.6. Let $a, b \in \mathbb{R}^2$. If the corresponding vectors $\vec{a}, \vec{b}$ both lie on the x -axis or both lie on the y -axis, then

$$\vec{a} \oplus \vec{b} = a + b$$

Proof. If $a = (a, 0)$ and $b = (b, 0)$, then $a + b = (a + b, 0)$. The corresponding plane vector sum lies on the x -axis and has signed length $a + b$. The argument is analogous for vectors lying on the y -axis. ■

Lemma 4.2.7. Let $a, b \in \mathbb{R}^2$. If the corresponding plane vector $\vec{a}$ lies on the x -axis and the corresponding plane vector $\vec{b}$ lies on the y -axis, then

$$\vec{a} \oplus \vec{b} = a + b$$

Proof. The premise implies $a = (a, 0)$ and $b = (0, b)$, and therefore $a + b = (a, b)$. By the definition of arrow addition using the parallelogram (in this case a rectangle), $\vec{a} \oplus \vec{b}$ is the vector corresponding to the point (a, b) . ■

Proposition 4.2.8 (Equivalence of Algebraic and Geometric Addition). For all $a, b \in \mathbb{R}^2$ and corresponding plane vectors $\vec{a}, \vec{b}$,

$$a + b = \vec{a} \oplus \vec{b}$$

Proof. Let $a = (a_1, a_2)$ and $b = (b_1, b_2)$. Their algebraic sum is $a + b = (a_1 + b_1, a_2 + b_2)$.

We can decompose a as $a = (a_1, 0) + (0, a_2)$. By Lemma 4.2.7, since the corresponding arrows lie on the x and y axes,

$$\vec{a} = (\widehat{a_1, 0}) \oplus (\widehat{0, a_2})$$

Similarly, $\vec{b} = (\widehat{b_1, 0}) \oplus (\widehat{0, b_2})$.

Thus, by commutativity and associativity of $\oplus$ (Propositions 4.1.6 and 4.1.7),

$$\vec{a} \oplus \vec{b} = [(\widehat{a_1, 0}) \oplus (\widehat{0, a_2})] \oplus [(\widehat{b_1, 0}) \oplus (\widehat{0, b_2})] = [(\widehat{a_1, 0}) \oplus (\widehat{b_1, 0})] \oplus [(\widehat{0, a_2}) \oplus (\widehat{0, b_2})]$$

By Lemma 4.2.6,

$$(\widehat{a_1, 0}) \oplus (\widehat{b_1, 0}) = (a_1, 0) + (b_1, 0) = (a_1 + b_1, 0)$$

and

$$(\widehat{0, a_2}) \oplus (\widehat{0, b_2}) = (0, a_2) + (0, b_2) = (0, a_2 + b_2)$$

Finally, by Lemma 4.2.7 again,

$$\vec{a} \oplus \vec{b} = (\widehat{a_1 + b_1, 0}) \oplus (\widehat{0, a_2 + b_2}) = (a_1 + b_1, 0) + (0, a_2 + b_2) = (a_1 + b_1, a_2 + b_2) = a + b$$

Note. The identification of $\vec{a} \oplus \vec{b}$ with $a + b$ gives us a way to do geometry algebraically without sketches and protractors. Now that we know they are the same, we don't need both $+$ and $\oplus$. We will use a and $a + b$ to denote both algebraic and geometric meanings, depending on context. For example, a can mean the 2-tuple $(a_1, a_2) \in \mathbb{R}^2$ or the plane vector with tail at the origin and head at the point (a_1, a_2) , and $a + b$ can mean either componentwise addition in $\mathbb{R}^2$ or parallelogram-rule addition in the plane.

Proposition 4.2.9 (Geometric Interpretation of Scalar Multiplication). Let $a \in \mathbb{R}^2$ and let $t \in \mathbb{R}$. For the plane vectors a and ta :

- ta lies on the same line as a
- ta is $|t|$ times longer than a
- ta is in the same direction as a if $t > 0$
- ta is in the opposite direction as a if $t < 0$

Proof. If $t = 0$, then $ta = 0$ is the zero vector, which lies on every line through the origin, has length zero, and is considered to point in every direction, satisfying the proposition trivially. Now assume $t \neq 0$.

Case 1: If a lies on the x -axis or y -axis, then $a = (a, 0)$ or $a = (0, a)$. Thus:

- $ta = (ta, 0)$ or $ta = (0, ta)$, in both cases lying on the same axis as a .
- The length of a is $|a|$ and the length of ta is $|ta| = |t||a|$.
- If $t > 0$, then $\text{sgn}(ta) = \text{sgn}(a)$ and the arrows are in the same direction.
- If $t < 0$, then $\text{sgn}(ta) = -\text{sgn}(a)$ and the arrows are in opposite directions.

Case 2: If a is not on the x -axis or y -axis, then $a = (a_1, a_2)$ with $a_1, a_2 \neq 0$, and

$$ta = t[(a_1, 0) + (0, a_2)] = (ta_1, 0) + (0, ta_2)$$

Consider the right triangle with vertices $O = (0, 0)$, $A = (a_1, 0)$, $B = (a_1, a_2)$, and the right triangle with vertices $O = (0, 0)$, $A' = (ta_1, 0)$, $B' = (ta_1, ta_2)$.

Since the ratios of lengths of corresponding legs are equal (both equal to $|t|$), the triangles are similar. Therefore:

- The hypotenuse angles are equal, so they lie on the same line.
- The ratio of hypotenuse lengths is also $|t|$, so $|ta| = |t||a|$.
- If $t > 0$, both triangles lie in the same quadrant, so ta and a point in the same direction.
- If $t < 0$, the triangles lie in opposite quadrants, so ta and a point in opposite directions.

Note. In looser but more helpful phrasing, Proposition 4.2.9 says that ta is the arrow obtained by stretching (or shrinking) the arrow a by a factor $|t|$, keeping it on the same line, in the same direction if $t > 0$ and reversing direction if $t < 0$.

Definition 4.3.1 (Linear Vector Space). Let F be a field. A **linear vector space over F** is a set V equipped with a binary operation $+$: $V \times V \rightarrow V$, called **addition**, and an operation $\cdot$: $F \times V \rightarrow V$, called **multiplication by a scalar**, that satisfy the following properties (called axioms of the space):

Axioms of addition:

- Closure:** $\forall u, w \in V, u + w \in V$
- Commutativity:** $\forall u, w \in V, u + w = w + u$
- Associativity:** $\forall u, w, v \in V, (u + w) + v = u + (w + v)$
- Additive identity:** $\exists 0 \in V$ such that $\forall u \in V, u + 0 = u$
- Additive inverse:** $\forall u \in V, \exists (-u) \in V$ such that $u + (-u) = 0$

Axioms of scalar multiplication:

- Closure:** $\forall u \in V$ and $\forall \lambda \in F, \lambda \cdot u \in V$
- Distributivity over vector addition:** $\forall u, w \in V$ and $\forall \lambda \in F, \lambda(u + w) = \lambda u + \lambda w$
- Distributivity over scalar addition:** $\forall u \in V$ and $\forall \lambda, \mu \in F, (\lambda + \mu)u = \lambda u + \mu u$
- Associativity-like property:** $\forall u \in V$ and $\forall \lambda, \mu \in F, (\lambda\mu)u = \lambda(\mu u)$
- Multiplicative identity:** $\forall u \in V$, if $\bar{1} \in F$ is the multiplicative identity, then $\bar{1} \cdot u = u$

Notation. • Members of V are called **vectors**; we use Latin underlined letters to denote vectors, e.g., $\underline{u}, \underline{w}$.

- Members of F are called **scalars**; we use Greek letters to denote scalars, e.g., λ, μ .
- We use the same symbol $+$ to denote addition in V and addition in F ; context distinguishes between the two operations.
- We use the same symbol $\cdot$ (or, more often, juxtaposition) to denote scalar multiplication and multiplication in F ; context distinguishes between the two operations.
- We denote the additive identity in F by $\bar{0}$ and the additive identity in V by 0 .
- Linear vector space** is often shortened to **linear space**, **vector space**, or simply **space**.

Note. The axioms of vector addition are the same as the axioms of addition in the field F . It follows that every property of addition between scalars that relied only on the addition axioms will hold for addition between vectors, in any vector space over any field.

Lecture 25: Immediate Properties of Vector Spaces

4.4 Immediate Properties

Note. We now list various properties that must hold for any vector space over any field. These properties have been seen before in other contexts (as properties of n -tuples, matrices, or numbers). The proofs utilize only the properties guaranteed by the axioms of the space, making them universally applicable.

Theorem 4.4.1 (Generalized Associativity). Let V be a vector space over the field F and let $u_1, \dots, u_n \in V$. Every possible parenthesis ordering of the expression

$$u_1 + u_2 + u_3 + \dots + u_n$$

is equivalent to the left-associated ordering:

$$(\dots((u_1 + u_2) + u_3) \dots + u_n)$$

Proof. This is a particular case of the general Theorem 0.3.7. ■

Theorem 4.4.2 (Generalized Distributivity). Let V be a vector space over the field F .

- For any $u \in V$ and any $\lambda_1, \dots, \lambda_n \in F$,

$$(\lambda_1 + \dots + \lambda_n)u = \lambda_1 u + \dots + \lambda_n u$$

- For any $\lambda \in F$ and any $u_1, \dots, u_n \in V$,

$$\lambda(u_1 + \dots + u_n) = \lambda u_1 + \dots + \lambda u_n$$

Proof. We prove both parts by induction.

Part i:

Base case ($n = 2$): $(\lambda_1 + \lambda_2)u = \lambda_1 u + \lambda_2 u$ is axiom viii.

Induction hypothesis: Suppose $(\lambda_1 + \dots + \lambda_k)u = \lambda_1 u + \dots + \lambda_k u$ for any k scalars.

Induction step:

$$\begin{aligned} (\lambda_1 + \dots + \lambda_k + \lambda_{k+1})u &= ((\lambda_1 + \dots + \lambda_k) + \lambda_{k+1})u = (\lambda_1 + \dots + \lambda_k)u + \lambda_{k+1}u \\ &= (\lambda_1 u + \dots + \lambda_k u) + \lambda_{k+1}u = \lambda_1 u + \dots + \lambda_k u + \lambda_{k+1}u \end{aligned}$$

Part ii:

Base case ($n = 2$): $\lambda(u_1 + u_2) = \lambda u_1 + \lambda u_2$ is axiom vii.

Induction hypothesis: Suppose $\lambda(u_1 + \dots + u_k) = \lambda u_1 + \dots + \lambda u_k$ for any k vectors.

Induction step:

$$\begin{aligned} \lambda(u_1 + \dots + u_k + u_{k+1}) &= \lambda((u_1 + \dots + u_k) + u_{k+1}) = \lambda(u_1 + \dots + u_k) + \lambda u_{k+1} \\ &= (\lambda u_1 + \dots + \lambda u_k) + \lambda u_{k+1} = \lambda u_1 + \dots + \lambda u_k + \lambda u_{k+1} \end{aligned}$$

Theorem 4.4.3 (Uniqueness). Let V be a vector space over the field F .

- The neutral vector in V (the zero vector, 0) is unique.
- The inverse of any $u \in V$ is unique.

Lecture 24: Spacedef

4.3 Definition/Examples

Proof. Part i: Suppose 0 and $0'$ are both neutral with respect to vector addition. Then

$$0 = 0 + 0' = 0'$$

where the first equality holds because $0'$ is neutral, and the second equality holds because 0 is neutral.

Part ii: Suppose u' and u'' are both inverses of u . Then

$$u' = u' + 0 = u' + (u + u'') = (u' + u) + u'' = 0 + u'' = u''$$

■

Theorem 4.4.4 (Properties of Zeros and Identity). Let V be a vector space over the field F .

- $\forall u \in V, u + u = u \Rightarrow u = 0$
- $\forall \lambda \in F, \lambda 0 = 0$
- $\forall u \in V, \tilde{0}u = 0$ (where $\tilde{0}$ is the additive identity in F)
- If $\lambda u = 0$, then $\lambda = \tilde{0}$ or $u = 0$

Proof. Part i: Suppose $u + u = u$. Then

$$u = u + 0 = u + (u + (-u)) = (u + u) + (-u) = u + (-u) = 0$$

Part ii: For any $\lambda \in F, \lambda 0 = \lambda(0 + 0) = \lambda 0 + \lambda 0$, so by part i, $\lambda 0 = 0$.

Part iii: For any $u \in V, \tilde{0}u = (\tilde{0} + \tilde{0})u = \tilde{0}u + \tilde{0}u$, so by part i, $\tilde{0}u = 0$.

Part iv: Suppose $\lambda u = 0$ and $\lambda \neq \tilde{0}$. Then λ has a multiplicative inverse λ^{-1} in F . Thus

$$u = \tilde{1}u = (\lambda^{-1}\lambda)u = \lambda^{-1}(\lambda u) = \lambda^{-1}0 = 0$$

Therefore, if $\lambda \neq \tilde{0}$, then $u = 0$. ■

Theorem 4.4.5 (Properties of Negation). Let V be a vector space over the field F .

- $\forall u \in V, (-\tilde{1})u = -u$ (where $-\tilde{1}$ is the additive inverse of $\tilde{1}$ in F)
- $\forall u \in V$ and $\forall \lambda \in F, (-\lambda)u = -(\lambda u)$
- $\forall u \in V$ and $\forall \lambda \in F, \lambda(-u) = -(\lambda u)$
- $\forall u \in V, -(-u) = u$
- $\forall u, w \in V, -(u + w) = (-u) + (-w)$

Proof. Part i: We show that $(-\tilde{1})u$ is the additive inverse of u :

$$u + (-\tilde{1})u = \tilde{1}u + (-\tilde{1})u = (\tilde{1} + (-\tilde{1}))u = \tilde{0}u = 0$$

By uniqueness of the additive inverse (Theorem 4.4.3.ii), $(-\tilde{1})u = -u$.

Part ii: For any $\lambda \in F$ and $u \in V$,

$$\lambda u + (-\lambda)u = (\lambda + (-\lambda))u = \tilde{0}u = 0$$

By uniqueness of the additive inverse, $(-\lambda)u = -(\lambda u)$.

Part iii: For any $\lambda \in F$ and $u \in V$,

$$\lambda u + \lambda(-u) = \lambda(u + (-u)) = \lambda 0 = 0$$

By uniqueness of the additive inverse, $\lambda(-u) = -(\lambda u)$.

Part iv: We have $u + (-u) = 0$ by axiom v. Adding $-(-u)$ to both sides:

$$\begin{aligned} (u + (-u)) + (-(-u)) &= 0 + (-(-u)) \\ u + ((-u) + (-(-u))) &= -(-u) \\ u + 0 &= -(-u), \quad u = -(-u) \end{aligned}$$

Part v: We show that $(-u) + (-w)$ is the additive inverse of $u + w$:

$$\begin{aligned} (u + w) + ((-u) + (-w)) &= (u + w) + ((-w) + (-u)) \\ &= ((u + w) + (-w)) + (-u) \\ &= (u + (w + (-w))) + (-u) = (u + 0) + (-u) \\ &= u + (-u) = 0 \end{aligned}$$

By uniqueness of the additive inverse, $-(u + w) = (-u) + (-w)$. ■

Definition 4.4.6 (Subtraction). Let V be a vector space over the field F . For all $u, w \in V$, the **difference** $u - w$ is the vector

$$u - w := u + (-w)$$

Note. The definition works because, by Theorem 4.4.3.ii, the inverse of w is unique.

Theorem 4.4.7 (Properties of Subtraction). Let V be a vector space over the field F . For all $u, v, w \in V$:

- $(u - v) + w = u - (v - w)$
- $(u - v) - w = u - (v + w)$
- $u - w = 0 \Leftrightarrow u = w$

Proof. Part i:

$$\begin{aligned} (u - v) + w &= (u + (-v)) + w = u + ((-v) + w) = u + (w + (-v)) \\ &= u + (w - v) = u + ((-(-v)) + (-w)) = u + (-(v + (-w))) \\ &= u + (-(v - w)) = u - (v - w) \end{aligned}$$

Part ii:

$$\begin{aligned} (u - v) - w &= (u + (-v)) - w = (u + (-v)) + (-w) \\ &= u + ((-v) + (-w)) = u + (-(v + w)) = u - (v + w) \end{aligned}$$

Part iii: ($\Rightarrow$) Suppose $u - w = 0$. Then

$$u = u + 0 = u + (w + (-w)) = (u + (-w)) + w = (u - w) + w = 0 + w = w$$

Conversely, ($\Leftarrow$) if $u = w$ then

$$u - w = u + (-w) = w + (-w) = 0$$

■

Lecture 26: Subspaces

4.5 Subspaces

Definition 4.5.1 (Subspace). A subset $W \subseteq V$ of a vector space V over a field F is called a **subspace** of V if it is a vector space over F using the addition and multiplication-by-a-scalar defined in V .

Note. When verifying that $W \subseteq V$ is a subspace, we need not check all ten space axioms. Since V is already a vector space, axioms ii (commutativity), iii (associativity of addition), vii, viii, ix (distributivity properties), and x (scalar identity) are automatically satisfied for any subset. We only need to verify axioms i (closure under addition), iv (existence of zero), v (existence of inverses), and vi (closure under scalar multiplication).

Furthermore, if W is a subspace of V , then the zero vector of W is the same as the zero vector of V , and the additive inverse of any $u \in W$ is the same in both W and V .

Theorem 4.5.2 (Subspace Test). A nonempty subset $W \subseteq V$ of a vector space V over a field F is a subspace of V if and only if for any $w_1, w_2 \in W$ and any $\lambda_1, \lambda_2 \in F$,

$$\lambda_1 w_1 + \lambda_2 w_2 \in W$$

Proof. We show that this single condition implies all four necessary axioms:

- Axiom i (closure under addition): Take $\lambda_1 = \lambda_2 = \tilde{1}$. Then for any $w_1, w_2 \in W$, we have $w_1 + w_2 = \tilde{1}w_1 + \tilde{1}w_2 \in W$.
- Axiom iv (existence of zero): Since $W \neq \emptyset$, there exists some $w \in W$. Take $\lambda_1 = \lambda_2 = \tilde{0}$. Then $\tilde{0}w + \tilde{0}w = 0 + 0 = 0 \in W$.
- Axiom v (existence of inverses): For any $w \in W$, take $\lambda_1 = -\tilde{1}$ and $\lambda_2 = \tilde{0}$. Then $(-\tilde{1})w + \tilde{0}w = -w + 0 = -w \in W$.
- Axiom vi (closure under scalar multiplication): For any $w \in W$ and $\lambda \in F$, take $\lambda_1 = \lambda$ and $\lambda_2 = \tilde{0}$. Then $\lambda w + \tilde{0}w = \lambda w + 0 = \lambda w \in W$.

Conversely, if W is a subspace, then it satisfies closure under both addition and scalar multiplication, so the condition holds. ■

Note. While Theorem 4.5.2 provides a compact way to express the subspace condition, in practice it is often clearer to verify the three conditions separately:

- $0 \in W$ (existence of zero)
- For all $w_1, w_2 \in W, w_1 + w_2 \in W$ (closure under addition)
- For all $w \in W$ and $\lambda \in F, \lambda w \in W$ (closure under scalar multiplication)

Note. Every vector space V has at least two subspaces, called the **trivial subspaces**:

- $W = \{0\}$ (the zero subspace)
- $W = V$ (the whole space)

Lecture 27: Spans

4.6 Spans

Definition 4.6.1 (Linear Combination). Let V be a vector space over a field F .

- A **linear combination** of vectors $v_1, \dots, v_n \in V$ is a sum of the form

$$\lambda_1 v_1 + \dots + \lambda_n v_n = \sum_{i=1}^n \lambda_i v_i$$

where $\lambda_1, \dots, \lambda_n \in F$.

- Given a subset $S \subseteq V$, if a vector $u \in V$ is equal to some linear combination of vectors in S , we say that u is **linearly dependent on** the set S .

Note. Properties of linear combinations:

- The scalars λ_i are called the **coefficients** of the linear combination.
- The number of vectors in a linear combination is arbitrary but finite.
- The coefficients λ_i may be zero; therefore 0 is linearly dependent on any nonempty set of vectors.
- If u is dependent on $S \subseteq V$, then it is also dependent on $S \cup W$ for any $W \subseteq V$ (additional vectors can be added with zero coefficients).

Definition 4.6.2 (Span). Let $K \subseteq V$ be a subset of a vector space V . The set of all possible linear combinations of vectors from K is called the **span** of K and denoted $\text{Sp}(K)$. More precisely,

$$\text{Sp}(K) = \left\{ \sum_{i=1}^n \lambda_i v_i : n \in \mathbb{N}, v_1, \dots, v_n \in K, \lambda_1, \dots, \lambda_n \in F \right\}$$

For completeness, we define $\text{Sp}(\emptyset) = \{0\}$.

Theorem 4.6.3 (Properties of Span). Let K be a nonempty subset of a vector space V over field F . Then:

- $\text{Sp}(K)$ is a subspace of V that contains K .
- If W is a subspace of V that contains K , then W also contains $\text{Sp}(K)$.

Proof. Part i: First, we show that $K \subseteq \text{Sp}(K)$. For any $w \in K$, we have $w = 1 \cdot w$, which is a linear combination of a single vector from K , so $w \in \text{Sp}(K)$. Thus $K \subseteq \text{Sp}(K)$.

Next, we show that $\text{Sp}(K)$ is a subspace using Theorem 4.5.2. Let $u, v \in \text{Sp}(K)$ and $\lambda, \mu \in F$. Then there exist:

- vectors $u_1, \dots, u_p \in K$ and scalars $\alpha_1, \dots, \alpha_p \in F$ such that $u = \alpha_1 u_1 + \dots + \alpha_p u_p$
- vectors $v_1, \dots, v_q \in K$ and scalars $\beta_1, \dots, \beta_q \in F$ such that $v = \beta_1 v_1 + \dots + \beta_q v_q$

Therefore,

$$\lambda u + \mu v = \lambda \alpha_1 u_1 + \dots + \lambda \alpha_p u_p + \mu \beta_1 v_1 + \dots + \mu \beta_q v_q \in \text{Sp}(K)$$

since this is a linear combination of vectors from K . By Theorem 4.5.2, $\text{Sp}(K)$ is a subspace.

Part ii: Let W be a subspace of V that contains K . Let $u \in \text{Sp}(K)$. Then there exist vectors $u_1, \dots, u_p \in K$ and scalars $\alpha_1, \dots, \alpha_p \in F$ such that

$$u = \alpha_1 u_1 + \dots + \alpha_p u_p$$

Since $K \subseteq W$, we have $u_i \in W$ for all i . Since W is a subspace, it is closed under scalar multiplication, so $\alpha_i u_i \in W$ for all i . Since W is closed under addition, $\alpha_1 u_1 + \dots + \alpha_p u_p \in W$. Therefore $u \in W$, which shows $\text{Sp}(K) \subseteq W$. ■

Theorem 4.6.4 (Uniqueness of Minimal Subspace). Let K be a nonempty subset of a vector space V and let W be a subspace of V that contains K and is contained in every subspace of V that contains K . Then $W = \text{Sp}(K)$.

Proof. We prove equality by showing mutual containment.

By Theorem 4.6.3.ii, since W is a subspace that contains K , we have $\text{Sp}(K) \subseteq W$.

By Theorem 4.6.3.i, $\text{Sp}(K)$ is a subspace that contains K . By the premise, W is contained in every subspace of V that contains K , so $W \subseteq \text{Sp}(K)$.

Therefore $W = \text{Sp}(K)$. ■

Note. Theorem 4.6.4 characterizes $\text{Sp}(K)$ as the *smallest* subspace containing K . That is, $\text{Sp}(K)$ is the unique minimal subspace of V containing K .

Theorem (Properties of Span Operations). Let V be a vector space.

- A subset $U \subseteq V$ is a subspace if and only if $\text{Sp}(U) = U$.
- For every $K \subseteq V$, $\text{Sp}(\text{Sp}(K)) = \text{Sp}(K)$.

Proof. Part i: ($\Rightarrow$) Suppose U is a subspace. By Theorem 4.6.3.i, $U \subseteq \text{Sp}(U)$. For the reverse inclusion, let $u \in \text{Sp}(U)$. Then $u = \lambda_1 u_1 + \dots + \lambda_n u_n$ for some $u_i \in U$ and $\lambda_i \in F$. Since U is a subspace, it is closed under scalar multiplication and addition, so $u \in U$. Thus $\text{Sp}(U) \subseteq U$, and therefore $\text{Sp}(U) = U$.

($\Leftarrow$) Suppose $\text{Sp}(U) = U$. By Theorem 4.6.3.i, $\text{Sp}(U)$ is a subspace, so U is a subspace.

Part ii: By Theorem 4.6.3.i, $\text{Sp}(K)$ is a subspace. By part i, $\text{Sp}(\text{Sp}(K)) = \text{Sp}(K)$. ■

Definition (Row Space). Let $A \in M_{m \times n}(F)$ and denote by $a_i \in F^n$ the vector whose components are the entries in the i -th row of A . The subspace $\text{Sp}(a_1, \dots, a_m)$ is called the **row space** of A .

Note. Row-equivalent matrices have the same row space because elementary row operations produce linear combinations of the original rows.

Theorem 4.6.5 (Equality of Spans). Let K, T be nonempty subsets of a vector space V over F . Then $\text{Sp}(K) = \text{Sp}(T)$ if and only if $K \subseteq \text{Sp}(T)$ and $T \subseteq \text{Sp}(K)$.

Proof. ($\Rightarrow$) Suppose $\text{Sp}(K) = \text{Sp}(T)$. By Theorem 4.6.3.i, $K \subseteq \text{Sp}(K)$ and $T \subseteq \text{Sp}(T)$. Therefore $K \subseteq \text{Sp}(K) = \text{Sp}(T)$ and $T \subseteq \text{Sp}(T) = \text{Sp}(K)$.

($\Leftarrow$) Suppose $K \subseteq \text{Sp}(T)$ and $T \subseteq \text{Sp}(K)$. By Theorem 4.6.3.ii:

- Since $K \subseteq \text{Sp}(T)$ and $\text{Sp}(T)$ is a subspace containing $\text{Sp}(T)$, we have $\text{Sp}(K) \subseteq \text{Sp}(\text{Sp}(T)) = \text{Sp}(T)$
- Since $T \subseteq \text{Sp}(K)$ and $\text{Sp}(K)$ is a subspace containing $\text{Sp}(K)$, we have $\text{Sp}(T) \subseteq \text{Sp}(\text{Sp}(K)) = \text{Sp}(K)$.

Since $\text{Sp}(K) \subseteq \text{Sp}(T)$ and $\text{Sp}(T) \subseteq \text{Sp}(K)$, we conclude $\text{Sp}(K) = \text{Sp}(T)$. ■

Lecture 28: Linear Independence

4.7 Linear Independence

Definition 4.7.1 (Linear Independence). Let V be a vector space over a field F and let K be a subset of V . K is called **linearly independent** if there exist distinct vectors $v_1, \dots, v_n \in K$ and scalars $\lambda_1, \dots, \lambda_n \in F$, not all zero, such that

$$\lambda_1 v_1 + \dots + \lambda_n v_n = 0$$

Note. Any set containing the zero vector cannot be independent, and the empty set is linearly independent (vacuously true).

Theorem 4.7.2 (Characterization of Linear Dependence). Let V be a vector space over F and let K be a subset of V containing at least two vectors. Then K is linearly dependent if and only if at least one of the vectors in K is equal to a linear combination of other vectors in K .

Proof. ($\Rightarrow$) Suppose K is linearly dependent. Then there exist distinct vectors $v_1, \dots, v_n \in K$ and scalars $\lambda_1, \dots, \lambda_n \in F$, not all zero, such that $\lambda_1 v_1 + \dots + \lambda_n v_n = 0$. WLOG, suppose $\lambda_1 \neq 0$. If $n = 1$, then $\lambda_1 v_1 = 0 \Rightarrow v_1 = 0$; take any $w \neq v_1 \in K$ and $v_1 = 0 \cdot w$ is the required linear combination. If $n > 1$, then $\lambda_1 v_1 = -\lambda_2 v_2 - \dots - \lambda_n v_n$, so $v_1 = \lambda_1^{-1}(-\lambda_2 v_2 + \dots + \lambda_n v_n)$ is the required linear combination.

($\Leftarrow$) Suppose $\exists w \in K$ that is a linear combination of other vectors in K . Then there exist n distinct vectors $v_1, \dots, v_n \in K$ and scalars $\lambda_1, \dots, \lambda_n$ such that $\lambda_1 v_1 + \dots + \lambda_n v_n = w$. Therefore $\lambda_1 v_1 + \dots + \lambda_n v_n + (-1)w = 0$. Since $-1 \neq 0$, this is a non-trivial linear combination equal to 0, so K is linearly dependent. ■

Lecture 29: The Space F^n

4.8 F^n

Theorem 4.8.1 (Linear Combinations as Systems of Equations). Given a field F and k vectors in F^n , a vector $b = (b_1, \dots, b_n) \in F^n$ is a linear combination of $a_1, \dots, a_k$ with coefficients $s_1, \dots, s_k$ if and only if the k -tuple $(s_1, \dots, s_k)$ is a solution of the linear system whose augmented coefficient matrix is

$$S = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1k} & b_1 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nk} & b_n \end{pmatrix}$$

where the columns of S are the vectors $a_1, \dots, a_k, b$.

Proposition 4.8.2 (Independence as Homogeneous Systems). The vectors $a_1, \dots, a_k \in F^n$ are linearly independent if and only if the homogeneous system of n equations in k unknowns whose coefficient matrix has $a_1, \dots, a_k$ as columns does not have a non-trivial solution.

Proof. This is an application of Theorem 4.8.1 to the case $b = 0$. ■

Theorem 4.8.3 (More Vectors Than Dimension Implies Dependence). Given $a_1, \dots, a_k \in F^n$, if $k > n$ then they are linearly dependent.

Proof. Let $a_1 = (a_{11}, a_{21}, \dots, a_{n1}), \dots, a_k = (a_{1k}, a_{2k}, \dots, a_{nk})$. By Proposition 4.8.2, the vectors are linearly dependent iff there is a non-trivial solution to the homogeneous system with the non-augmented coefficient matrix from Theorem 4.8.1. This is a homogeneous system of n equations in k unknowns. By Theorem 2.6.2, if $k > n$ the system has a non-trivial solution. ■

Definition 4.8.4 (Standard Basis Vectors). In F^n , the vectors

$$\begin{aligned} e_1 &= (1, 0, 0, \dots, 0) \\ e_i &= (0, 0, \dots, 1, \dots, 0) \\ e_n &= (0, 0, \dots, 1) \end{aligned}$$

are called the **standard basis vectors** of F^n . We use the same notation e_i in different spaces; for example, in $\mathbb{R}^4$ we write $e_2 = (0, 1, 0, 0)$ while in $\mathbb{R}^6$ we write $e_2 = (0, 1, 0, 0, 0, 0)$.

Theorem 4.8.5 (Standard Basis Vectors are Independent). The n standard basis vectors in F^n are linearly independent.

Proof. Consider the equation $s_1 e_1 + \dots + s_n e_n = 0$. This expands to $(s_1, 0, \dots, 0) + (0, s_2, \dots, 0) + \dots + (0, 0, \dots, s_n) = (0, 0, \dots, 0)$, which gives $(s_1, s_2, \dots, s_n) = (0, 0, \dots, 0)$. Thus $s_i = 0$ for all i , so the only solution is trivial. ■

Definition 4.8.6 (Spanning Set). Let V be a vector space. Vectors $a_1, \dots, a_k \in V$ **span** V if $\text{Sp}(a_1, \dots, a_k) = V$. For $a_1, \dots, a_k \in F^n$, they span F^n if $\forall b \in F^n, \exists s_i \in F$ such that $\sum_{i=1}^k s_i a_i = b$.

Theorem 4.8.7 (Standard Basis Vectors Span F^n). The standard basis vectors of F^n span F^n .

Proof. For any $b = (b_1, \dots, b_n) \in F^n$,

$$b_1 e_1 + \dots + b_n e_n = (b_1, 0, \dots, 0) + \dots + (0, 0, \dots, b_n) = (b_1, b_2, \dots, b_n) = b$$

Lemma 4.8.8 (Dependent Linear Combinations). Let $a_1, \dots, a_k \in F^n$ be any k vectors and let $b_1, \dots, b_m \in F^n$ be m vectors that are each a linear combination of the a vectors. If $m > k$, then the vectors $b_1, \dots, b_m$ are not independent.

Proof. Let $b_i = \sum_{j=1}^k x_{ji} a_j$. Then for any $s_1, \dots, s_m \in F$,

$$\sum_{i=1}^m s_i b_i = \sum_{i=1}^m s_i \sum_{j=1}^k x_{ji} a_j = \sum_{j=1}^k \left(\sum_{i=1}^m s_i x_{ji} \right) a_j$$

A non-trivial solution to the homogeneous system (known x_{ji} , unknown s_i) will provide a non-trivial linear combination $\sum_{i=1}^m s_i b_i = 0$. When $m > k$, a non-trivial solution exists by Theorem 2.6.2. ■

Theorem 4.8.9 (Fewer Than n Cannot Span). Let $a_1, \dots, a_k \in F^n$. If $k < n$, then the vectors do not span F^n .

Proof. Suppose $a_1, \dots, a_k$ span F^n and $k < n$. Then every vector $b \in F^n$ is a linear combination of the a vectors. In particular, $e_1, \dots, e_n$ are linear combinations of the a vectors. Therefore there are n vectors that are each a linear combination of the k vectors $a_1, \dots, a_k$. By Lemma 4.8.8, $e_1, \dots, e_n$ are not linearly independent. But by Theorem 4.8.5, they are. Contradiction. Therefore $a_1, \dots, a_k$ do not span F^n . ■

Conclusion. By Theorem 4.8.9, if $k < n$ then $a_1, \dots, a_k$ do not span (F^n) . By Theorem 4.8.3, if $k > n$ then $a_1, \dots, a_k$ are not independent. Therefore, a set of vectors that is independent and spans F^n has exactly n vectors.

Chapter 5

Bases

Lecture 30: Reminders

- Lemma 5.0.1 (Various Truths).**
1. If $K = \{\underline{v}\}$ then K is dependent iff $\underline{v} = 0$.
 2. If K is independent and $T \subseteq K$, then T is independent.
 3. If K is dependent and $T \supseteq K$, then T is dependent.
 4. If K is independent and $\underline{v} \notin K$ is not dependent on K , then $K \cup \{\underline{v}\}$ is independent.
 5. If K is independent and $\underline{v} \notin K$ is dependent on K , then $K \cup \{\underline{v}\}$ is dependent.
 6. If $\text{Sp}(K \cup \{\underline{v}\}) = \text{Sp}(K)$, then $\underline{v}$ is dependent on K .
 7. If $\underline{v}$ is dependent on K , then $\text{Sp}(K \cup \{\underline{v}\}) = \text{Sp}(K)$.

Lecture 31: Bases and Dimensions

5.1 Bases and Dimensions

Definition 5.1.1 (Basis). Let V be a vector space over field F . A set $B \subseteq V$ is called a **basis** of V if B is linearly independent and B spans V (i.e., $\text{Sp}(B) = V$).

Definition 5.1.2 (Finitely Generated). A vector space V is **finitely generated** if there exists a finite set T such that $\text{Sp}(T) = V$.

Note. Infinite sets of vectors spanning spaces leads to complications making theorems more cumbersome, so we restrict ourselves to finitely generated spaces.

Theorem 5.1.3 (Existence of Finite Basis). Every finitely-generated vector space V has a finite basis.

Proof. Let B be a set of n vectors that spans V and assume without loss of generality that $\underline{0} \notin B$.

If $n = 0$, then $B = \emptyset$ and is independent and therefore a basis (this is the trivial case where $V = \{\underline{0}\}$, from Definition 4.6.2).

If $n = 1$, then $B = \{\underline{v}\}$ is independent (Lemma 5.0.1) and therefore a basis.

If B is linearly independent, then it is a basis. Otherwise, there exists a vector $\underline{v}_1 \in B$ that is a linear combination of the other vectors in B . The set $B_1 = B - \{\underline{v}_1\}$ includes $n - 1$ vectors and, since $\underline{v}_1$ is dependent on B_1 , we have $\text{Sp}(B_1 \cup \{\underline{v}_1\}) = \text{Sp}(B_1) = V$ (Lemma 5.0.1.viii).

If B_1 is independent, then it is a basis. Otherwise, there exists $\underline{v}_2$ that is dependent on the set $B_2 = B_1 - \{\underline{v}_2\}$ with $n - 2$ vectors, and $\text{Sp}(B_2) = V$.

After at most $n - 1$ such steps, we will find a finite set that is both spanning and independent, and therefore a basis. ■

Theorem 5.1.4 (Uniqueness of Basis Representation). Let V be a vector space over field F and let $B = \{\underline{u}_1, \dots, \underline{u}_n\}$ be a basis of V . Then every $\underline{w} \in V$ is a unique linear combination of the basis vectors.

Proof. Let $\underline{w} \in V$. Since $\text{Sp}(B) = V$, there exist $\lambda_1, \dots, \lambda_n \in F$ such that $\underline{w} = \sum_{i=1}^n \lambda_i \underline{u}_i$.

Suppose there is another combination with coefficients $\mu_1, \dots, \mu_n \in F$ such that $\underline{w} = \sum_{i=1}^n \mu_i \underline{u}_i$. Then $\underline{0} = \underline{w} - \underline{w} = \sum_{i=1}^n \lambda_i \underline{u}_i - \sum_{i=1}^n \mu_i \underline{u}_i = \sum_{i=1}^n (\lambda_i - \mu_i) \underline{u}_i$. Since B is a basis, it is independent. Therefore, for all $i \in \{1, \dots, n\}$, we have $\lambda_i - \mu_i = 0$. Hence $\lambda_i = \mu_i$ for all $i \in \{1, \dots, n\}$. ■

Lecture 32: Dimension of a Vector Space

5.2 Dimension of a Vector Space

Lemma 5.2.1 (Steinitz Exchange Lemma). Let $A = \{\underline{v}_1, \dots, \underline{v}_k\}$ and $B = \{\underline{w}_1, \dots, \underline{w}_m\}$ be subsets of a vector space V over field F . If A is a spanning set and B is independent, then $m \leq k$.

Proof. We will show that if $m > k$, there is a non-trivial linear combination $\sum_{i=1}^m x_i \underline{w}_i = \underline{0}$. Since $\text{Sp}(A) = V$, there exist $\lambda_{ji} \in F$ such that for all $\underline{w}_i \in B$, $\underline{w}_i = \sum_{j=1}^k \lambda_{ji} \underline{v}_j$. A linear combination of vectors from B is of the form $\sum_{i=1}^m x_i \underline{w}_i = \sum_{i=1}^m x_i \sum_{j=1}^k \lambda_{ji} \underline{v}_j = \sum_{j=1}^k (\sum_{i=1}^m \lambda_{ji} x_i) \underline{v}_j$. If $m > k$, the homogeneous system has a non-trivial solution (by Theorem 2.6.2), meaning there exist $x_i \in F$, not all zero, such that $\sum_{i=1}^m \lambda_{ji} x_i = 0$ for all $j = 1, \dots, k$. This makes $\underline{0}$ a non-trivial linear combination of B vectors, meaning B is not independent. Therefore, if B is independent, $m \leq k$. ■

Theorem 5.2.2 (Properties of n -Vector Bases). Let V be a vector space over field F . If V has a basis with n vectors, then:

- (i) Any set in V with more than n vectors is dependent
- (ii) Any set in V with fewer than n vectors does not span V
- (iii) Any independent set of n vectors in V is a spanning set
- (iv) Any spanning set of V with n vectors is independent
- (v) Every basis of V has exactly n vectors

Proof. Let $B = \{\underline{v}_1, \dots, \underline{v}_n\}$ be a basis of V , and let $K = \{\underline{w}_1, \dots, \underline{w}_k\} \subseteq V$.

- (i) Since B is a spanning set, by Lemma 5.2.1, if $k > n$ then K is dependent.
- (ii) If $k < n$ and K is a spanning set, by Lemma 5.2.1, B would not be independent, which is a contradiction.
- (iii) If $k = n$ and K is independent and $\text{Sp}(K) \neq V$, there is a $\underline{u} \notin \text{Sp}(K)$ not dependent on K , and $K \cup \{\underline{u}\}$ would be an independent set (Lemma 5.0.1.v) with $n + 1$ vectors, in contradiction to part (i).
- (iv) If $k = n$ and $\text{Sp}(K) = V$ and K is not independent, there exists $\underline{u} \in K$ that is a linear combination of the other vectors in K , and the set $K - \{\underline{u}\}$ would be a spanning set (Lemma 5.0.1.vii) with $n - 1$ vectors, in contradiction to part (ii).
- (v) Being independent, a basis must have no more than n vectors (by part (i)); being a spanning set it must have at least n vectors (by part (ii)). ■

Definition 5.2.3 (Dimension). Let V be a finitely-generated vector space. The **dimension** of V is the number of vectors in a basis of V , denoted $\dim V$. For completeness, we define $\dim\{\underline{0}\} = 0$.

Theorem 5.2.4 (Completion to a Basis). If $A = \{\underline{v}_1, \dots, \underline{v}_k\}$ is an independent set in an n -dimensional space V , there exist vectors $\underline{w}_{k+1}, \dots, \underline{w}_n \in V$ such that $B = \{\underline{v}_1, \dots, \underline{v}_k, \underline{w}_{k+1}, \dots, \underline{w}_n\}$ is a basis of V . In other words, any linearly independent set in an n -dimensional space can be completed to a basis.

Lecture 33: Coordinates and the Great Isomorphism

5.3 Coordinates and the Great Isomorphism

Definition 5.3.0 (Ordered Basis). A sequence $\underline{v}_1, \dots, \underline{v}_n$ of vectors in V is called an **ordered basis** if it is independent and spanning.

Note. Note that while $\{\underline{v}, \underline{w}, \underline{u}\}$ and $\{\underline{w}, \underline{u}, \underline{v}\}$ are the same set (and thus the same basis), the sequences $\underline{v}, \underline{w}, \underline{u}$ and $\underline{w}, \underline{u}, \underline{v}$ are not the same ordered basis, since $\alpha \underline{v} + \beta \underline{w} + \gamma \underline{u} \neq \alpha \underline{w} + \beta \underline{u} + \gamma \underline{v}$.

Definition 5.3.1 (Coordinates). Let $B = \{\underline{v}_1, \dots, \underline{v}_n\}$ be a basis of the vector space V over F and let $\underline{v} \in V$. The scalars $\alpha_1, \dots, \alpha_n \in F$ appearing in the linear combination

$$\underline{v} = \alpha_1 \underline{v}_1 + \dots + \alpha_n \underline{v}_n$$

are called the **coordinates** of $\underline{v}$ under the (ordered) basis B . The n -tuple $(\alpha_1, \dots, \alpha_n) \in F^n$ is called the **coordinate vector** of $\underline{v}$ under the basis

B and denoted $[\underline{v}]_B$. The column vector $\begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}$ is also called the coordinate vector of $\underline{v}$ under the basis B and denoted $[\underline{v}]_B$.

Theorem 5.3.2 (The Coordinate Isomorphism). Let V be a vector space over F and let $B = \{\underline{v}_1, \dots, \underline{v}_n\}$ be a basis of V . The mapping $f: V \rightarrow F^n$ that matches every vector $\underline{v} \in V$ with its coordinate vector $[\underline{v}]_B \in F^n$ is:

- (i) One-to-one: $[\underline{w}]_B = [\underline{w}']_B \Rightarrow \underline{w} = \underline{w}'$
- (ii) Onto: $\forall \underline{u} \in F^n, \exists \underline{v} \in V$ such that $[\underline{v}]_B = \underline{u}$
- (iii) Preserves vector addition: $\forall \underline{w}, \underline{w}' \in V, [\underline{w}]_B + [\underline{w}']_B = [\underline{w} + \underline{w}']_B$
- (iv) Preserves scalar multiplication: $\forall \underline{w} \in V$ and $\forall \lambda \in F, \lambda [\underline{w}]_B = [\lambda \underline{w}]_B$

Proof. (i) Let $\underline{u} = (u_1, \dots, u_n) = [\underline{w}]_B$ and $\underline{u}' = (u'_1, \dots, u'_n) = [\underline{w}']_B$. If $\underline{u} = \underline{u}'$, then $u_i = u'_i$ for all $i \in \{1, \dots, n\}$, and $\underline{w} = \sum_{i=1}^n u_i \underline{v}_i = \sum_{i=1}^n u'_i \underline{v}_i = \underline{w}'$.

(ii) Let $\underline{u} = (u_1, \dots, u_n) \in F^n$. Then $\underline{w} = \sum_{i=1}^n u_i \underline{v}_i \in V$ and by definition $[\underline{w}]_B = \underline{u}$.

(iii) Let $\underline{u} = (u_1, \dots, u_n) = [\underline{w}]_B$ and $\underline{u}' = (u'_1, \dots, u'_n) = [\underline{w}']_B$. Then $[\underline{w} + \underline{w}']_B = [\sum_{i=1}^n u_i \underline{v}_i + \sum_{i=1}^n u'_i \underline{v}_i]_B = [\sum_{i=1}^n (u_i + u'_i) \underline{v}_i]_B = (u_1 + u'_1, \dots, u_n + u'_n) = \underline{u} + \underline{u}' = [\underline{w}]_B + [\underline{w}']_B$.

(iv) Let $\underline{u} = (u_1, \dots, u_n) = [\underline{w}]_B$ and let $\lambda \in F$. Then $[\lambda \underline{w}]_B = [\lambda \sum_{i=1}^n u_i \underline{v}_i]_B = [\sum_{i=1}^n \lambda u_i \underline{v}_i]_B = (\lambda u_1, \dots, \lambda u_n) = \lambda \underline{u} = \lambda [\underline{w}]_B$. ■

Definition 5.3.3 (Isomorphism). A mapping from one vector space to another that is one-to-one, onto, and preserves vector addition and scalar multiplication is called an **isomorphism**.

Theorem 5.3.4 (Properties of Coordinate Vectors). Let B be a basis of the vector space V over field F . Then:

- (i) $[\underline{v}]_B = \underline{0} \in F^n$ if and only if $\underline{v} = \underline{0} \in V$
- (ii) $[-\underline{v}]_B = -[\underline{v}]_B$
- (iii) $[\sum_{i=1}^k \lambda_i \underline{v}_i]_B = \sum_{i=1}^k \lambda_i [\underline{v}_i]_B$ for any k vectors $\underline{v}_1, \dots, \underline{v}_k$
- (iv) $\{\underline{v}_1, \dots, \underline{v}_k\} \subseteq V$ is independent if and only if $\{[\underline{v}_1]_B, \dots, [\underline{v}_k]_B\} \subseteq F^n$ is independent

Proof. Parts (i), (ii), and (iii) follow directly from Theorem 5.3.2.

(iv) We prove the contrapositive in both directions.

First direction: If $\{\underline{v}_1, \dots, \underline{v}_k\}$ are dependent, there exist $\lambda_1, \dots, \lambda_k \in F$, not all zero, such that $\sum_{i=1}^k \lambda_i \underline{v}_i = \underline{0} \in V$. By parts (i) and (iii), $\underline{0} \in F^n = [\underline{0} \in V]_B = [\sum_{i=1}^k \lambda_i \underline{v}_i]_B = \sum_{i=1}^k \lambda_i [\underline{v}_i]_B$ is a non-trivial linear combination of $\underline{0} \in F^n$, proving that $\{[\underline{v}_i]_B\}$ are dependent.

Second: If $\{[\underline{v}_1]_B, \dots, [\underline{v}_k]_B\}$ are dependent, there exist $\lambda_1, \dots, \lambda_k \in F$, not all zero, such that $\sum_{i=1}^k \lambda_i [\underline{v}_i]_B = \underline{0} \in F^n$. By parts (i) and (iii), $[\underline{0} \in V]_B = \underline{0} \in F^n = \sum_{i=1}^k \lambda_i [\underline{v}_i]_B = [\sum_{i=1}^k \lambda_i \underline{v}_i]_B$ and thus by Theorem 5.3.2(i), $\underline{0} \in V = \sum_{i=1}^k \lambda_i \underline{v}_i$ is a non-trivial linear combination, proving that $\{\underline{v}_i\}$ are dependent. ■